IncuVision: A Smart Chicken Egg Incubator with Automated Environmental Control and YOLOv8-Integrated Multi-Egg Candling System

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Poultry plays a vital role in food security and agricultural sustainability in the Philippines. It provides meat and eggs to the world population which is the rich source of the vitamins and proteins that play a great role in health development (Khan et al., 2021). It is a vital source of revenue for the entire nation and has grown to a significant economic component, accounting for one-third of the nation's total agricultural output (Briones & Espineli, 2023). The poultry industry is not only the most dynamic and fast-growing branch of livestock farming, but it also has the most intensive production methods (Efremova, 2018).

Within this intensive production framework, fertility and hatchability are primary metrics of success. Fertility refers to the percentage of incubated eggs that are fertile while hatchability is the percentage of fertile eggs that hatch. Therefore, it's critical to comprehend the variables affecting egg hatchability and fertility.

While identifying fertile eggs is a crucial first step, maintaining their viability throughout the 21-day cycle depends entirely on the stability of the incubation environment. Traditional incubation relies heavily on the precise regulation of external factors, primarily temperature and humidity, which serve as the physiological stimulus for embryo growth. However, Onagbesan et al. (2007) emphasizes that atmospheric composition, specifically the balance of Carbon Dioxide (CO_2) and Oxygen (O_2), is equally vital. Proper gas exchange is necessary to support the metabolic demands of the developing embryo; insufficient ventilation

can lead to CO_2 toxicity or O_2 starvation, both of which significantly increase embryonic death and decrease final hatchability rates.

To manage these variables, one of the most essential processes in incubation management is egg candling, a traditional method used to determine egg fertility by shining light through the eggshell to observe embryo development. It is used to examine the clear eggs or dead embryos for hatchery managers to use in maintaining quality and analyzing poor hatches (Ernst et al., 2004). However, manual candling is labor-intensive, subjective, and prone to human error, especially in large-scale hatchery operations. Inconsistent evaluation may result in delayed removal of infertile eggs, which can negatively affect incubation conditions and overall hatchability rates.

The impact of such inconsistencies is significant, as the hatchability of eggs is a vital economic factor for hatcheries and breeder farms, directly influencing production efficiency and overall profitability. It is affected by various factors, including breeder flock health, fertility levels, and proper egg handling and management practices. The early detection of non-hatchable eggs allows hatchery operators to identify potential problems at an early stage, thereby saving incubation space, reducing labor and handling costs, and improving resource utilization. Studies report that hatchery success rates typically range from 85% to 95%, (Ecc DM, 2025) indicating that a portion of eggs still fail to hatch due to undetected fertility or developmental issues. It can be expressed either relative to the total number of eggs set, in which case it encompasses both fertility and embryo survival, or relative to the number of fertile eggs (determined by candling) (Wolc et al., 2010).

Addressing these developmental issues, computer vision for egg embryo identification begins with the image acquisition stage, which can be performed using various methods and tools. Different acquisition techniques have been utilized, including the use of thermal cameras,

hyperspectral imaging systems, and standard digital cameras (Saifulla et al., 2021). Advancements in computer vision and artificial intelligence have enabled more accurate and automated methods for analyzing visual patterns inside eggs during candling. Computer Vision can enhance image clarity, detect blood vessels, and classify fertile versus infertile eggs with improved precision and consistency.

Despite the widespread use of conventional egg incubators in poultry production, many of these systems are already capable of maintaining basic environmental conditions such as temperature and humidity with reasonable accuracy. In some cases, these traditional incubators can adequately support embryo development, which raises the need to clearly distinguish the added value of more advanced systems. Chicken breeders typically monitor and control egg incubators manually and on a periodic basis to ensure proper hatching conditions. However, this practice is time-consuming and inefficient in terms of labor and energy (Sila et al., 2018). These limitations highlight the need for a more comprehensive and intelligent incubation system that not only maintains optimal environmental conditions but also enhances monitoring accuracy, reduces human intervention, and improves overall hatchability outcomes.

While advancements in computer vision and sensor technology offer potential solutions, there remains a lack of integrated, fully automated systems that combine internal visual classification with real-time atmospheric monitoring in a single incubation unit. Considering these factors, the researchers will design and develop a fully automated and monitored system egg incubator integrated with a built-in multi-egg candling system. The device incorporates YOLOv8 to classify egg fertility accurately by analyzing internal visual features, to reduce the reliance on manual inspection. In addition, CO_2 (carbon dioxide) and O_2 (oxygen) sensors are embedded within the system to continuously monitor gas concentrations inside the incubator, ensuring a balanced and optimal environment for embryo development.

By integrating automation and modern technologies, this study aims to contribute to the modernization of poultry farming practices, improve hatchability rates, offer a sustainable and cost-efficient solution adaptable to different farming scales, and establish a foundation for future innovations in poultry production systems.

1.2 Statement of the Problem

The primary objective of this research is to develop and evaluate IncuVision, a smart chicken egg incubator integrated with automated environmental control and a YOLOv8-based multi-egg candling system for enhanced egg fertility monitoring and incubation performance. This study aims to automate both environmental regulation and fertility detection to reduce manual intervention, improve consistency in monitoring, and support more efficient incubation management. By integrating computer vision and sensor-based control within a single system, the research seeks to provide a reliable and practical solution adaptable to poultry farming operations.

Specifically, this study aims to address the following:

1. To design and implement an **automated system** that regulates temperature, humidity, and gas concentrations required for the chicken **egg incubation process.**
2. To develop a specialized image dataset for egg candling by capturing and labeling high-resolution images of chicken embryos within the developed incubator system.
3. To train a **YOLOv8 fertility detection** model using a labeled dataset of egg candling images for accurate classification of **chicken egg fertility** (fertile or infertile).
4. To integrate the trained YOLOv8 model for automated multi-egg candling into a **real-time monitoring system** for environmental parameters and egg fertility.

5. To evaluate the effectiveness of the developed system in terms of egg fertility detection and incubation performance compared to traditional methods.

1.3. Significance of the Study

This study focuses on the application of system automation to reduce manual intervention and improve the efficiency and reliability of egg incubation. It features a multi-egg candling system with computer vision using machine learning for accurate fertility detection. The system also integrates CO_2 and O_2 sensors to monitor internal gas levels and maintain optimal conditions for embryo development. Additionally, WiFi enables real-time monitoring, tracking, and data logging. Its portable design ensures flexibility for use in poultry farms and research settings.

The findings of this study are anticipated to offer advantages for different sectors, such as:

1. **Poultry Farmers.** It will reduce the manual labor of the farmers and more efficient incubation management through automated monitoring and fertility detection.
2. **Department of Agriculture.** It is to improve agricultural productivity by introducing innovative and technology-driven incubation solutions for the poultry sector.
3. **Students.** Gain knowledge and practical insights through the integration of engineering, automation and artificial intelligence in real-world agricultural technologies.
4. **Future Researchers.** It can serve as reference and foundation for further research on smart incubation systems, image processing, and IoT-based agricultural technologies.

1.4 Scope and Delimitation

This study focuses on developing IncuVision, a portable egg incubator designed to support mobile poultry farmers by significantly reducing manual intervention. The primary goal is to offer flexibility and convenience in chicken egg incubation through a compact and transportable design that aligns with the dynamic workflows of modern farming.

A key feature of IncuVision is its innovative automated monitoring and control system, which regulates critical incubation parameters such as temperature, humidity, and periodic egg turning. This is achieved through embedded sensors and actuators that read and adjust these variables in real time to maintain the optimal atmosphere required for healthy embryo development. Distinctively, the system incorporates CO_2 and O_2 sensors to monitor internal gas concentrations, ensuring proper ventilation throughout the incubation cycle.

The incubator also features an integrated multi-egg candling system limited to a 56-egg capacity that utilizes computer vision and machine learning to accurately detect egg fertility without disturbing the eggs. This automated classification helps lessen the manual labor required from poultry farmers and eliminates the subjectivity and inconsistency often associated with traditional manual candling processes.

Furthermore, the device is equipped with WiFi functionalities to monitor and track the entire incubation process through a dedicated mobile application. The system features seamless data logging, achieved through a localized wireless network. This allows for real-time data synchronization between the incubation hardware and the mobile interface without the requirement of an active internet connection, ensuring operational continuity even in remote agricultural areas with limited telecommunications infrastructure. The user interface is designed to be intuitive and user-friendly, catering to both farmers and researchers with varying levels of technical knowledge.

1.5. Definition of Terms

Automation: The integration of technology and control systems to perform incubation tasks with minimal human intervention, aimed at enhancing operational consistency.

Candling: The process of using a concentrated light source to observe internal egg development; in this study, it refers to the automated classification of fertility.

Egg Incubation: The method of maintaining fertile eggs within strictly regulated environmental conditions, specifically temperature, humidity, and gas levels (CO_2 and O_2) to ensure healthy embryo development.

IncuVision: The specific prototype of a portable, smart egg incubator developed in this research, featuring integrated multi-egg candling and atmospheric monitoring.

Machine Learning (ML): A subset of artificial intelligence used to train the system's algorithm to automatically distinguish between fertile, infertile, and non-developing eggs based on a collected dataset.

You Only Look Once v8(YOLOv8): A state-of-the-art computer vision model used for real-time object detection. In this study, it automates the classification of eggs into fertile or infertile categories by analyzing visual features from candling images.

CHAPTER 2

REVIEW OF LITERATURE

This chapter highlights the key concepts and theories that align with the problem and methodologies outlined in this study. It presents a range of related literature, previous studies, and related work from other researchers in the same field. These references form the foundation for understanding and supporting the research's central aspects.

2.1. Related Literature

2.1.1. Egg Incubation Process

Egg incubation involves maintaining fertilized eggs under ideal environmental conditions to support embryo development until hatching, typically taking 21 days for chickens and up to 30 days for other poultry species (Burgess, 2023). While originally done using broody hens, the practice has largely shifted to mechanical systems that precisely control conditions, significantly boosting efficiency and success rates. This advancement has greatly improved poultry farming in the Philippines, a vital source of food and livelihood, enabling farmers to achieve hatchability rates exceeding 80% compared to the less reliable outcomes of traditional methods (Meghan, 2017).

2.1.1.1 Temperature and Humidity

Successful egg incubation depends on controlled conditions, with temperature and humidity being the most critical factors. Based on Philippine standards, chicken eggs require temperatures between 37.22°C and 38.9°C, ideally maintained around 37.5°C, while humidity should be kept at 58% for the first 18 days and increased to 66-75% during the final days to facilitate hatching

(Boleli et al., 2016; Noiva et al., 2020). Deviations from these levels can be harmful: low temperatures slow development and reduce hatch rates, while excessive heat causes deformities and death. Similarly, insufficient humidity leads to moisture loss and large air cells, whereas too much moisture interferes with nutrient absorption and results in sticky chicks that struggle to emerge (Shrack et al., n.d.; Linden, 2014). Because of their significant impact, standard values for temperature and humidity must be strictly followed, as summarized in the table below.

Table 2.1. *Standard Temperature and Humidity Values (Philippines National Standard 2023)*

Type	**Setting Period (Days 1-18)**		**Hatching Period (Days 19-21)**	
	Relative humidity (%)	**Temperature (°C)**	**Relative humidity (%)**	**Temperature (°C)**
Chicken Egg	58	37.5	66-75	36.9

2.1.1.2 Ventilation

Adequate ventilation and air circulation are essential for successful incubation, directly influencing hatch rates and chick quality (Tona, 2022). Proper airflow supplies oxygen while removing carbon dioxide and excess moisture, as CO_2 levels above 0.5% can reduce incubation success and become harmful beyond 1.0%. Maintaining consistent temperatures throughout the incubator also prevents uneven heat distribution, which can reach differences of up to 7°C in still-air systems (Brinsea, 2019). Effective ventilation regulates humidity within the ideal range of 60–65%, ensuring eggs neither dry out nor accumulate excessive moisture. By implementing proper air exchange and forced-air systems, farmers can improve hatchability and produce healthier

chicks, making ventilation a critical factor in optimal incubation practices (Meijerhof, 2017).

Oxygen concentration is another important factor affecting embryo development during incubation. Under normal conditions, oxygen levels are maintained at 21% throughout incubation. During the setting phase (Days 1–18), carbon dioxide is maintained between 0.3% and 0.5% to support healthy embryo growth. As embryos enter the hatching phase (Days 19–21), acceptable CO_2 levels increase to 0.75% due to higher metabolic activity, while oxygen must remain at 21% to avoid reduced hatchability (Mauldin, 2008). Studies have shown that even a small decrease in oxygen concentration can negatively affect embryo development. Nangsuay et al. (2021) reported that reducing oxygen from 21% to 17% during incubation resulted in lower embryonic heat production, decreased body mass, and poorer chick development. Similarly, Agbehadzi et al. (2024) found that reduced oxygen exposure increased embryonic mortality and lowered hatchability rates. These findings indicate that every percentage drop in oxygen concentration can significantly affect embryo growth, survival, and chick quality, emphasizing the importance of maintaining stable oxygen levels inside the incubator.

Table 2.2 *Optimal Gas Conditions for Egg Incubation*

Phase	Carbon Dioxide	Oxygen
Settings (Day 1-18)	0.3% - 0.5%	21%
Hatching (Day 19-21)	0.75%	21%

2.1.1.3 Turning and Tilting

Alongside environmental controls like ventilation, regularly moving the eggs is vital to keep embryos from attaching to the inner shell membranes. This task requires turning the eggs at least four to six times each day until the final three days before hatching (Oliveira, 2020). A common technique involves tilting the eggs at a 45-degree angle in alternating directions every hour, which helps ensure efficient gas exchange and uniform nutrient distribution, ultimately leading to better development and higher hatchability (El Ashram, 2020). By following these procedures and using modern equipment, farmers can significantly improve results, produce healthier chicks, and support more sustainable agricultural practices.

2.1.2. Chicken Egg Fertility

Egg fertility refers to the ability to distinguish between fertilized and unfertilized eggs based on observable characteristics that emerge during the early stages of development. This distinction is fundamental in poultry production, as only fertile eggs have the potential to develop into viable embryos. Identifying fertility accurately allows hatchery managers to make informed decisions that directly influence productivity, resource allocation, and overall operational efficiency. Proper identification reduces unnecessary incubation of non-viable eggs, thereby saving time, energy, and costs while improving hatch success rates (Saifullah et al., 2021).

A comprehensive understanding of egg fertility also involves examining the biological processes underlying egg formation, structure, and fertilization. The roles of the hen's reproductive organs, along with the composition of the egg such as the yolk,

albumen, and shell are essential in supporting embryo development. These components provide nutrients, protection, and the appropriate environment for growth once fertilization occurs. Establishing proper incubation conditions and ensuring successful embryo development form the scientific basis for effective hatchery management practices (Kamrul et al., 2022).

Moreover, advancements in fertility detection show that embryo viability can be assessed during the early stages of incubation through observable physical and developmental changes. Detecting these changes, particularly within the first week, enables early identification of viable and non-viable eggs, which is critical for minimizing losses and optimizing hatchery operations. Zhu et al. (2024) emphasized the importance of timely and accurate fertility assessment, as it enhances efficiency, reduces resource waste, and improves overall production outcomes. One widely used method for fertility assessment is egg candling, which uses light to examine the internal condition of eggs during incubation. Through candling, visible indicators such as blood vessels, embryo movement, and darkened internal structures can help determine embryo viability and fertility status.

LED-based candling systems also play an important role in improving egg fertility detection. LEDs provide stable illumination, low heat emission, and clearer visibility of internal egg structures, making them effective for non-invasive fertility assessment. Proper LED color and light intensity are essential for accurately observing embryo development indicators such as blood vessels and darkened structures. Combining red and warm white (2700K) LEDs achieved 94.9% fertility detection accuracy on day 7 of incubation (Zhu et al., 2017). In terms of light intensity, a study conducted by Landicho et al. (2023) showed 32 LED modules with 300 lumens each

for each egg provided sufficient illumination for automated fertility detection. These findings show that optimized LED color and lumen intensity enhance the accuracy and reliability of egg candling systems (Landicho et al.,2023).

2.1.3. System Automation

Embryo development is fundamentally dependent on precise environmental control, establishing the scientific basis for automated incubation systems. Building on these principles, multiple researches demonstrate that utilizing microcontrollers to regulate temperature, humidity, and egg rotation significantly enhances system efficiency and reliability, enabling real-time management of conditions and preventing operational failures (Boleli et al., 2016). This body of work identifies a critical gap in current systems, particularly the limited integration of remote monitoring and control capabilities using Internet of Things (IoT) technology, which represents a key opportunity to further improve efficiency and minimize losses in hatchery operations (Kocher et al., 2022). Incubation outcomes are influenced not only by temperature and humidity but also by lighting conditions. Variations in light exposure, such as intermittent or green light, can directly affect embryo development, growth patterns, stress responses, and post-hatching adaptability (Yalcin et al., 2022).

In the context of fertility assessment, advanced non-destructive technologies such as hyperspectral imaging, machine vision, spectroscopy, and thermal imaging have proven highly effective compared to traditional methods. These techniques play a vital role in improving detection accuracy, highlighting the importance of high-quality data and robust models for developing automated systems that can reliably distinguish fertile eggs (Adegbenjo et al., 2020).

Additionally, the inherent limitations of conventional incubation systems, noting that manual control often leads to unstable environmental conditions and lower hatch rates. Automated control systems utilizing sensors and advanced control methodologies address these issues by providing consistent regulation, which improves stability and increases hatching success, making the entire incubation process more reliable and scientifically grounded (Zakaria, Diky et al., 2023).

2.1.4. YOLO-Based Egg Fertility Detection

Deep learning models significantly enhance the accuracy and efficiency of automated egg fertility detection, particularly for native Philippine chickens (Flores et al., 2026). This technology addresses the limitations of manual candling by optimizing parameters to achieve more precise results, demonstrating how artificial intelligence, specifically YOLO architectures, can improve agricultural processes and support real-time monitoring systems in livestock management. Further establish the effectiveness of YOLO as a real-time object detection model, noting that CNN-based architectures such as Mask R-CNN and other variants offer distinct advantages over traditional methods in terms of accuracy, processing speed, and scalability for automated fertility assessment (Pooja et al., 2025)

The integration of advanced imaging techniques with YOLO models has proven highly effective for non-invasive detection. Victor Massaki Nakaguchi et al., (2023) explored the combination of thermal imaging with YOLOv4 and YOLOv5, achieving high accuracy in identifying fertile and infertile eggs during early incubation stages compared to other models like SSD-MobileNet. Similarly, Tofael Ahamed et al., (2022) demonstrated that YOLO-based systems can effectively classify eggs at early phases,

although performance may vary depending on image quality and dataset size, highlighting the importance of proper data collection and model optimization.

Hybrid and optimized architectures continue to push the boundaries of detection capabilities. A system combining YOLOv8 with PatchCore, enabling precise localization and non-destructive detection of fertility and embryo anomalies using smartphone-based imaging (Thios et al., 2025). Youfu Liu et al., (2025) introduced AFF-YOLOX, a lightweight model designed to deliver fast and accurate results, while Li Xinyao et al. (2025) showed that modifying YOLO architectures with additional layers and attention mechanisms significantly improves feature extraction and overall performance.

Practical implementation is further supported by studies demonstrating integration with hardware. A development of portable system using YOLOv5s integrated with Raspberry Pi and cameras, enabling fast, real-time, and highly accurate detection that outperforms manual candling. This confirms that YOLO-based systems are not only technically effective but also practical and scalable for deployment in actual poultry farming operations (Ramos, Jed Edwin et al., 2025).

2.1.5. Real-time Monitoring System

Intelligent sensing based on commercial WiFi devices has emerged as a promising approach for next-generation monitoring systems due to its cost-effectiveness and widespread availability. According to Yao et al. (2022), this technology enables non-contact sensing and supports applications such as activity recognition and localization, thereby enhancing the efficiency of remote observation. Recent advancements have facilitated the development of low-cost sensor systems

capable of delivering near-real-time data with minimal deployment complexity, enabling high-frequency data acquisition essential for detecting environmental variations. WiFi-enabled networks offer significant potential for real-time monitoring and control, with reliable wireless communication identified as a critical factor for continuous system performance (Spinazzè et al., 2022).

These principles are directly relevant to the development of a smart incubation system. WiFi technology functions as the primary communication medium, enabling efficient transmission of environmental data, including temperature and humidity, in near real time. Its affordability and ease of implementation support the creation of an effective monitoring framework capable of maintaining precise environmental conditions. Moreover, the integration of wireless communication with cost-efficient computing platforms, such as the Raspberry Pi, enhances system functionality by enabling automated monitoring and control, thereby contributing to improved efficiency in incubation management (Jolles, 2021).

2.2. Related Studies

2.2.1. Egg Incubation Process

Research on egg incubation systems has led to various innovative developments aimed at addressing common challenges in the poultry industry. Significant studies from local and international contexts highlight these advancements, detailing both the methodologies employed and the issues that emerged during their investigations.

In a study by Peña et al. (2023), the incubator design utilized a proportional-integral (PI) controller combined with an Arduino Uno microcontroller and a regulated light bulb to maintain a target temperature of 37°C. The system also featured a display

module that showed real-time readings of temperature, humidity, and setpoint values. Proteus software was applied to simulate the electrical behavior of key components such as dimmers and optocouplers, while black-box modeling was used to define the system transfer function. In a separate development, Zakaria (2024) implemented a proportional-integral-derivative (PID) control mechanism to adjust the power supplied to an incandescent lamp, successfully stabilizing the operating temperature at 37°C. Nizarudeen et al. (2025) adopted an XHM452 controller, maintaining the internal temperature at 37.5°C, using a 40W bulb as the heat source and a DC fan to circulate warm air and achieve uniform heat distribution.

Aside from temperature, humidity management is also vital to incubation outcomes. The system described by Nizarudeen et al. (2025) maintained humidity between 70% and 85% through a humidifier paired with a moisture sensor. The device automatically adjusted water levels or activated spraying functions when necessary, while data was displayed in real time via an LCD screen. Pawar et al. (2026) expanded monitoring capabilities by incorporating relay-based heating modules, servo motors for automated egg turning, and an OLED display. Their design also supported remote data access and control through Wi-Fi integration with cloud services such as Blynk and ThingSpeak.

Gas concentrations within the incubator environment significantly affect hatchability and chick health. Fares et al. (2023) noted that standard incubation conditions generally maintain oxygen at 21% and carbon dioxide at 0.5%. However, modifications to these levels can influence results. According to Vieira et al. (2024), moderate increases in carbon dioxide (0.1% to 0.5%) during the early incubation phase improved hatch rates, while slightly higher levels (0.7% to 1%) later in the process

reduced the incidence of ascites syndrome. Supplementing oxygen, particularly in high-altitude environments, was also shown to enhance overall incubation success. Juarez-Estrada et al. (2024) warned that excessively high carbon dioxide concentrations (approximately 1.4%) could negatively affect embryo survival, especially in early developmental stages, due to toxic effects that exceed embryonic tolerance thresholds. The developmental phase of modern incubators often involves intricate design to integrate advanced functionalities. For example, Chiong et al. (2025) focused on designing an incubator that incorporates multi-egg candling, automated temperature and humidity regulation, and Bluetooth Low Energy (BLE) technology for seamless monitoring.

2.2.2. Chicken Egg Fertility

Research into egg fertility detection has become increasingly vital as the poultry industry seeks to improve operational efficiency and minimize economic losses. According to a report by Agriculture and Agri-Food Canada in 2017, more than 1.0 billion hatching eggs were produced in that year. With a fertility rate of about 82%, nearly 180 million eggs ultimately failed to hatch, representing an economic loss of over 300 million Canadian dollars. If these non-viable eggs could be identified early—ideally before being placed in incubators—they could still be repurposed as commercial table eggs or as raw materials for the food industry (Adegbenjo et al., 2020). On a global scale, approximately 130 million eggs are incubated annually, with infertility rates varying between 5% and 20% based on regional factors and hatchery practices. In 2023, the United States produced 1.31 billion eggs intended for hatching, accounting for 13% of the country's total egg production, yet roughly 7.5% of these were infertile. Beyond financial implications, undetected infertile eggs may decompose inside incubators,

583 releasing gases and potentially causing explosions, which can compromise hygiene,
584 waste energy and space, and increase disease risks (Ahmed et al., 2025).

585 Standard hatchery protocols typically involve collecting and storing eggs for up
586 to seven days before incubation begins. During preparation, eggs are carefully inspected
587 to remove those with cracks, irregular shapes, or dirt deposits, and selected eggs are
588 measured and weighed. Usually, medium-sized eggs are preferred and incubated under
589 controlled conditions—maintained at approximately 37.5°C and 60–65% relative
590 humidity—with automated turning performed every hour. Traditional methods involve
591 candling, a process where eggs are held against a light source to observe internal
592 development, conducted on the seventh and fourteenth days to separate infertile eggs
593 and identify deceased embryos (Sabales, 2024). To enhance the accuracy and efficiency
594 of these methods, researchers have turned to technology-based solutions. Fadchar and
595 Dela Cruz (2020) developed an imaging system that could capture images of five-day-
596 old eggs without damaging the shells. Out of 150 collected images, 100 were processed
597 through color segmentation to extract key parameters and used to train an artificial
598 neural network in MATLAB. The resulting model achieved a training accuracy of
599 98.6%, validation accuracy of 93.3%, and testing accuracy of 93.3%, giving it an
600 overall performance rate of 97%—a significant improvement over manual assessment.

601 In other research, Saifullah and Suryotomo (2021, 2022) explored two different
602 approaches using image datasets containing an equal number of fertile and infertile
603 eggs. In one method, image processing techniques were combined with
604 Backpropagation Neural Networks (BP-NN) and first-order statistical features. In
605 another experiment, they utilized a Support Vector Machine (SVM) classifier based on
606 five statistical measures: mean, entropy, variance, skewness, and kurtosis. The latter

method produced an identification success rate of 84.57%, proving that statistical feature extraction can serve as a reliable basis for automated classification. Çevik et al. (2022) took this integration of hardware and software a step further by designing a complete system that could be installed directly within incubators. The setup included a 1024 × 768 resolution camera controlled by a Raspberry Pi 4 unit, along with 10 W LED lighting positioned underneath the egg trays. This configuration allows continuous image capture from above without disturbing the incubation environment, creating the potential for real-time automated monitoring throughout the entire process.

2.2.3. System Automation

Research and development efforts in egg incubation technology focus on achieving precise environmental control and reliable fertility detection, both of which are essential to improve hatchability and operational efficiency. Optimal temperature management remains one of the most critical factors during the incubation period. It is recognized that temperature requirements can vary depending on the developmental stage and type of poultry. For instance, chicken eggs generally require 21 days of incubation, split into two distinct phases from day 1–18 and day 19–21, with different temperature settings needed to suit the growing embryo (Popović & Popović, 2024). To meet these requirements, Hadoune et al. (2021) developed a control system utilizing AVR microcontrollers and a PID algorithm to operate relays connected to heating plates and cooling fans, thereby stabilizing internal conditions. In another design, Arwan (2025) utilized an Arduino Uno-based setup that regulates temperature using a 12W bulb and 12V DC fan to maintain a steady 37.5°C, while humidity levels are managed using a moisture sensor and water pump system. In a local development, Mariani et al. (2021) described a system where a DHT11 sensor connected to an Arduino Uno

captures environmental data and triggers relays to activate incandescent bulbs when heating is required, with real-time values displayed on an LCD screen.

Alongside environmental regulation, modern incubation systems increasingly integrate methods to identify fertile eggs at an early stage. Traditional candling has been enhanced with imaging technologies and machine learning to make the process more objective and efficient. Tolentino et al. (2022) designed a four-layer incubation chamber equipped with built-in candling stations using fluorescent bulbs and DHT11 sensors connected to a Raspberry Pi 4 controller, enabling both environmental control and fertility screening within the same unit. Moving further into automated analysis, Musara et al. (2024) proposed an AI-based system that processes captured images by extracting features such as shape, color, texture, and internal structures like blood vessels. These features are then fed into trained machine learning models to classify eggs as fertile or infertile, with the results displayed on an integrated LCD screen. Nakaguchi and Ahamed (2022) explored the use of thermal imaging for this purpose, employing a compact FLIR® VUE™ 336 micro camera capable of capturing heat signatures in the 7.5–13.5 μm spectral range. Mounted above the incubator trays, this device collects radiometric data which can then be analyzed using deep learning algorithms to differentiate between fertilized and unfertilized eggs based on their thermal patterns.

2.2.4.YOLO-Based Egg Fertility Detection

Recent advancements in deep learning and computer vision have paved the way for more accurate and efficient egg fertility detection systems, addressing the limitations of traditional manual methods. Several studies have focused on adapting and optimizing the "You Only Look Once" (YOLO) algorithm for this purpose. Thios

and Zainuddin (2025) introduced a hybrid framework combining YOLOv8 with PatchCore, designed to detect developmental anomalies in chicken embryos. In this two-stage method, YOLOv8 first identifies the region of interest within images captured via smartphone candling, while PatchCore then analyzes that region to spot subtle irregularities such as blood rings or arrested development. The system achieved an AUROC of 0.96 and an overall accuracy of 89.10%, outperforming a standalone PatchCore model. Similarly, Aritha et al. (2025) applied YOLOv8 on a dataset of 4,396 chicken egg images, training the model over 100 epochs using Google Collaboratory. The results showed a precision of 93.2%, mean average precision of 98.5%, and recall of 87.2%, with particularly strong performance in identifying fertile eggs at 100% precision. In another technical development, Zhao et al. (2026) utilized the multi-scale feature fusion structure inherent to YOLOv8, which combines top-down and bottom-up pathways to integrate information at different levels, thereby enhancing the model's ability to detect objects across various scales.

To support these advanced models, researchers have also established robust training environments and datasets. Li et al. (2023) described a high-performance hardware setup including an Intel i9-12900K CPU, 64 GB RAM, and NVIDIA GeForce RTX 3090Ti GPU, running on a Windows 10 system with specialized software tools such as CUDA 11.7, Cudnn 8.6, and PyTorch 1.12, all operated using Python 3.8. Meanwhile, Mulani et al. (2024) demonstrated the effectiveness of YOLO compared to other detection algorithms like Faster R-CNN and SSD, reporting a precision of 96%, recall of 95.3%, and an overall accuracy of 97% in identifying DNA irregularities, making it both faster and more reliable than previous techniques. In a local study, Flores (2026) constructed a dataset consisting of 977 egg images supplemented with 47 background images to help the model learn non-object patterns and reduce false

positives. The data were split by session rather than randomly, ensuring that the model was tested on images taken under different conditions and on different days, thus preventing overfitting and providing a more realistic assessment of generalization capability.

2.2.5. Real-Time Monitoring System

The integration of Internet of Things (IoT) technology has revolutionized egg incubation by enabling efficient remote monitoring systems through wireless connectivity. Nugraha et al. (2024) emphasized that connecting incubators to the internet allows farmers to supervise the device from anywhere, utilizing specialized software to receive critical data such as temperature, humidity, and hatching status directly through mobile applications like Telegram. Similarly, Alawad and Ahmed (2021) developed a system where environmental parameters are sensed by the DHT11 module, displayed locally, and simultaneously transmitted to the Blynk application using an ESP8266 Wi-Fi module, ensuring real-time updates are accessible to users.

The reliability of data transmission is a key aspect of these systems, as highlighted by Afandi et al. (2022), who stated that gateways successfully send data through Wi-Fi networks to IoT platforms, allowing for accurate recording and analysis of temperature control characteristics. In support of this, Azamir et al. (2021) utilized an Arduino UNO microcontroller paired with an ESP8266 Wi-Fi module to transfer data to a database, presenting the information visually through graphs on websites and smartphone applications for comprehensive monitoring.

Furthermore, advanced configurations utilize Wi-Fi to connect various hardware components efficiently. Bhuiyin et al. (2025) explained that central control

units like the MKR 1010 MCU use wireless communication to publish data to MQTT brokers and Firebase, enabling remote access and control. Locally, in the Philippines, Saludo et al. (2025) developed the Intelli-Egg system, which integrates microcontrollers and sensors with Wi-Fi capabilities to provide automated environmental control and remote access via a web application. Additionally, Lahoylahoy et al. (2024) also explored the implementation of Wi-Fi technology in incubation setups, focusing on how wireless connectivity is established and utilized within the system to facilitate monitoring and data gathering. Collectively, these studies demonstrate that Wi-Fi-based monitoring systems provide a robust and convenient method for managing incubation parameters effectively.

2.3. Synthesis and Justification

IncuVision seeks to revolutionize the poultry industry by solving issues relating to environmental control and fertilization detection through incorporation of advanced technology such as automated processes, WiFi connection, and YOLO-based detection in vision technologies.

While many studies incorporate temperature and humidity into their scope of environmental regulation in the poultry incubators, there is little attention paid to monitoring Oxygen (O_2) and Carbon Dioxide (CO_2) levels in these systems. This becomes critical since embryos need oxygen for breathing purposes. IncuVision solves the problem by offering an integrated solution to monitor and balance O_2/CO_2 levels within the incubator. The incorporation of IoT within agriculture systems is highly encouraged due to its significant advantages (Kocher et al., 2022).

Moreover, multi-egg candling using effective technology has not been widely considered in past studies where manual candling and inefficient imaging techniques have been applied for this purpose. A fully integrated system incorporating all the mentioned aspects in a

single egg incubator hasn't been reported anywhere yet in any available literature. However, IncuVision incorporates YOLO-based detection technology that enables fast and simultaneous egg detection without interfering with them. This technique contributes to better hatchability rates as recent research supports this statement (Flores et al., 2026; Ramos et al., 2025).

Moreover, since IncuVision fully automates its processes, human errors are reduced, and the environmental conditions and turning of eggs are automatically controlled. The elimination of human errors and guesswork makes the process efficient since the environment is maintained at an optimal level (Boleli et al., 2016; Zakaria et al., 2023). IncuVision provides a more scientific method that ensures higher hatch rates and sustainable farming techniques.

2.4 Theoretical Framework

The theoretical framework for IncuVision is based on the principles of environmental control, system automation, wireless connectivity, and computer vision to optimize hatchability and efficiency. Proper management of parameters such as temperature, humidity, oxygen, and carbon dioxide is critical, as these factors directly influence embryo development and survival rates. Maintaining these conditions within optimal ranges ensures proper growth and prevents mortality, highlighting the necessity for precise regulation throughout the incubation process (Noiva et al., 2020; Tona et al., 2022).

To achieve this level of precision, the system utilizes automation technologies that minimize human intervention. Automated control systems ensure that environmental conditions remain stable and consistent, which is essential for improving hatchability and overall performance (Zakaria et al., 2023). Furthermore, the integration of Wi-Fi connectivity enables real-time data transmission and remote monitoring, addressing the limitations of traditional systems that lack remote accessibility. This feature allows farmers to monitor and

adjust conditions efficiently, aligning with the advancements in Internet of Things (IoT) technology (Kocher et al., 2022).

Additionally, the system employs YOLO-based detection to accurately identify fertile eggs. This method provides a fast, non-invasive, and highly accurate way of assessing egg fertility, which is vital for resource management and production efficiency. The effectiveness of this technology in classifying eggs and improving hatchery operations has been validated by recent studies, demonstrating its practical application in poultry farming (Flores et al., 2026; Ramos et al., 2025).

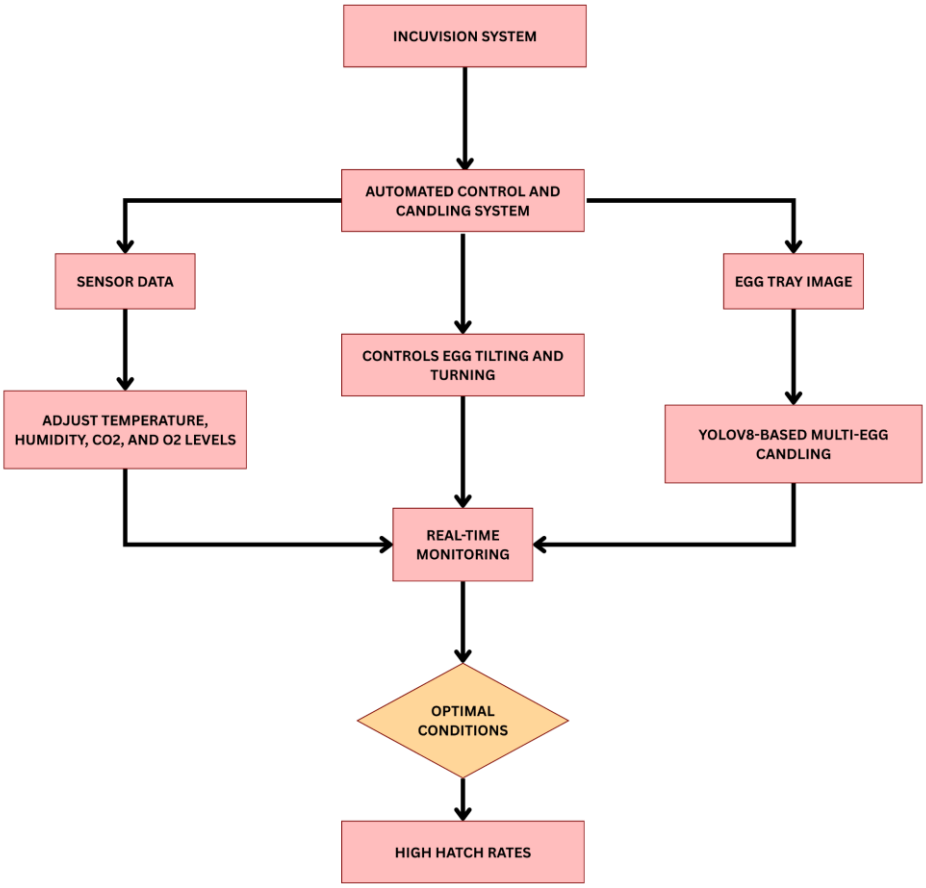


Figure 2.1. *Theoretical Framework*

761 The theoretical framework of the study is presented in Figure 2.1, and it emphasizes the
762 intelligent incubation system using automated control and monitoring and the YOLOv8
763 candling system that will provide optimal conditions for achieving higher hatching success
764 rates.
765

CHAPTER 3

METHODOLOGY

This chapter presents the methods that will be employed to design and evaluate IncuVision, a smart egg incubator that features automated environment control together with a YOLOv8-integrated multi-egg candling system. The automated control system is intended to help in maintaining key incubation conditions by monitoring temperatures, humidity, CO_2, and O_2 levels at various stages of the incubation process while also carrying out real-time egg fertility analysis using the YOLOv8 algorithm. Through the integration of the multi-parameter sensing, image-based analysis, and automated control mechanisms, the system enables reliable incubation conditions and accurate fertility assessment. This approach ensures reduced manual intervention, improved monitoring consistency, and provides a basis for evaluating the system's effectiveness in egg fertility detection and overall incubation performance.

3.1 Research Design

This study will employ a developmental and experimental research approach to design, implement, and evaluate a smart egg incubator system integrated with automated environmental control, gas sensing, and a YOLOv8-based vision system for egg fertility detection. The developmental phase will involve the design and construction of the incubator, incorporating temperature, humidity, carbon dioxide (CO_2), and oxygen (O_2) monitoring, along with a trained YOLOv8 model using labeled candling images. The experimental phase will focus on testing and validating the system's performance in maintaining optimal environmental stability across different incubation phases for optimal hatching conditions and fertility detection accuracy compared to traditional methods.

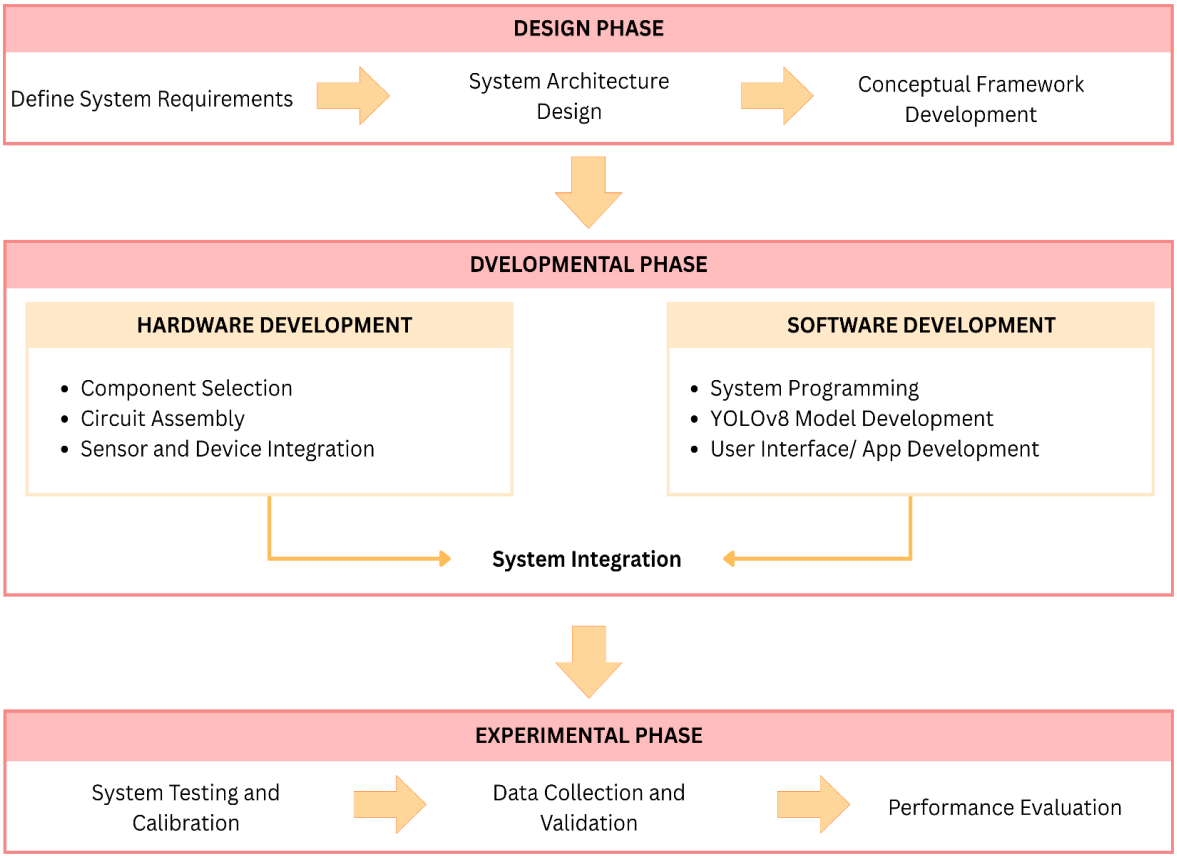


Figure 3.1. *General Phases of the Study*

Figure 3.1 presents a visual illustration of the study's research process, showing the sequential phases involved starting from the design of the system to its development and experimentation. It summarizes the entire process that will be followed in the development of the smart egg incubator system, including conceptualization, construction, and evaluation through integrated hardware and software development, followed by experimental testing to assess system performance and effectiveness.

3.1.1 Project Design Theory

The design theory of IncuVision is based on the integration of four main operational processes: precision environmental control, mechanical egg turning, automated vision-based fertility detection, and IoT-based remote monitoring. The

system is built to handle these operations automatically to ensure the best possible conditions for hatching.

The first process focuses on maintaining stable environmental conditions through a sensor-based monitoring system and a closed-loop control mechanism that regulates temperature and gas levels within the incubator. In parallel, the mechanical process ensures proper egg positioning through an automated turning system that supports consistent development.

Another key feature is the automated egg-candling process, which performs periodic image capture and analysis to assess egg fertility conditions using a trained YOLOv8 vision-based model. Lastly, the connectivity process enables data transmission from the system to a remote platform, allowing users to monitor conditions and results in real time through a mobile interface.

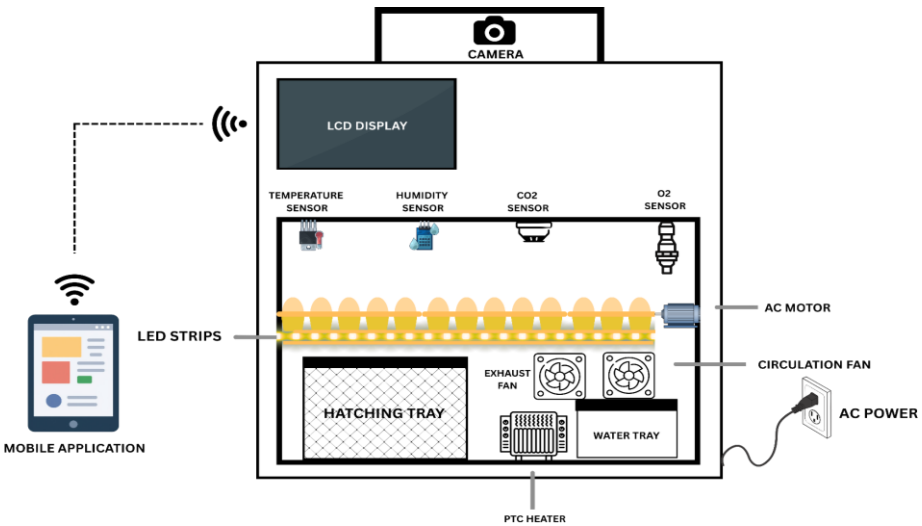


Figure 3.2. *Project Design Theory*

Figure 3.2 illustrates the project design theory of the system, highlighting the integration of the sensors, actuators, and vision components within the incubator to enable it to perform the operations stated above. It shows the placement of the

monitoring system, mechanical turning mechanism, environmental control units, and user interface modules. The configuration reflects the integration of sensing, actuation, and connectivity functions to support automated operation and real-time monitoring of the system.

3.2 Conceptual Framework

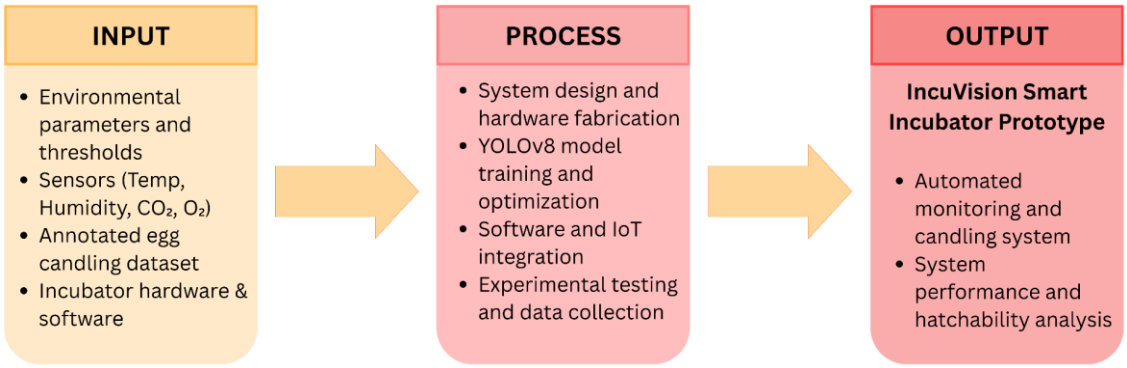


Figure 3.3. *Input-Process-Output (IPO) Model*

The conceptual framework of this study is depicted using the Input-Process-Output (IPO) model illustrated in Figure 3.3, highlighting the progression from inputs of the system, processes, to the output.

The Input covers the important inputs required for the study, including environmental standards and thresholds, sensors, annotated egg candling images data for machine learning model training, and minimum hardware and software requirements. The Process comprises the design and manufacturing of the incubator, training and fine-tuning of the YOLOv8 object detection model for egg fertility detection, software and Internet of Things connectivity, and experimentation on the prototype while incubating eggs. The Output comprises the findings of the study, including the smart incubator prototype designed, developed, and tested with its performance evaluation and hatchability.

3.3 Research Locale

This study will be conducted at the Electronics Engineering facilities of Western Mindanao State University, the researchers' residence, and a selected poultry farm in Zamboanga City. The university and the researcher's residence will serve as the primary sites for system development, including hardware assembly, sensor calibration, and software integration of the IncuVision smart egg incubator.

The experimental phase will take place at a selected farm in Zamboanga City, where the incubation trials and specimen sourcing will be conducted. This location is chosen to ensure that eggs are incubated shortly after collection, minimizing exposure to temperature fluctuations and handling stress. The farm environment also provides practical conditions for evaluating the system's performance in real-world poultry operations. By collaborating with local stakeholders, this study aims to support the adoption of automated incubation technologies, contributing to improved hatchability, efficiency, and sustainability in small-scale poultry farming within the region.

3.4. Respondents of the Study

The respondents of this study will include poultry farm owners and personnels engaged in egg incubation processes at the chosen research location. These participants will share details on existing incubation techniques, environmental monitoring systems, fertility evaluation methods, and difficulties faced during egg incubation by completing a pre-study farm assessment questionnaire before the experiments begin.

An experienced poultry practitioner will also participate as a respondent and serve as an expert validator. The expert validator will carry out manual candling evaluations and offer technical advice during the study's implementation. The fertility classifications given by the expert validator will act as the ground truth reference for assessing the performance of the YOLOv8-based fertility detection system incorporated into the created IncuVision Smart Egg Incubator. This study will utilize a blend of Convenience Sampling and Purposive Sampling methods.

Convenience Sampling. Farm owners and personnel available at the selected research site will be included as respondents due to their direct involvement in egg incubation activities and accessibility during the conduct of the study.

Purposive Sampling. The expert poultry practitioner will be selected based on their experience in egg incubation, manual candling procedures, and poultry management practices. The expert respondent will provide manual fertility classifications and technical validation of the data collected.

3.5 System Process Flow

3.5.1 Overall System Process Flow

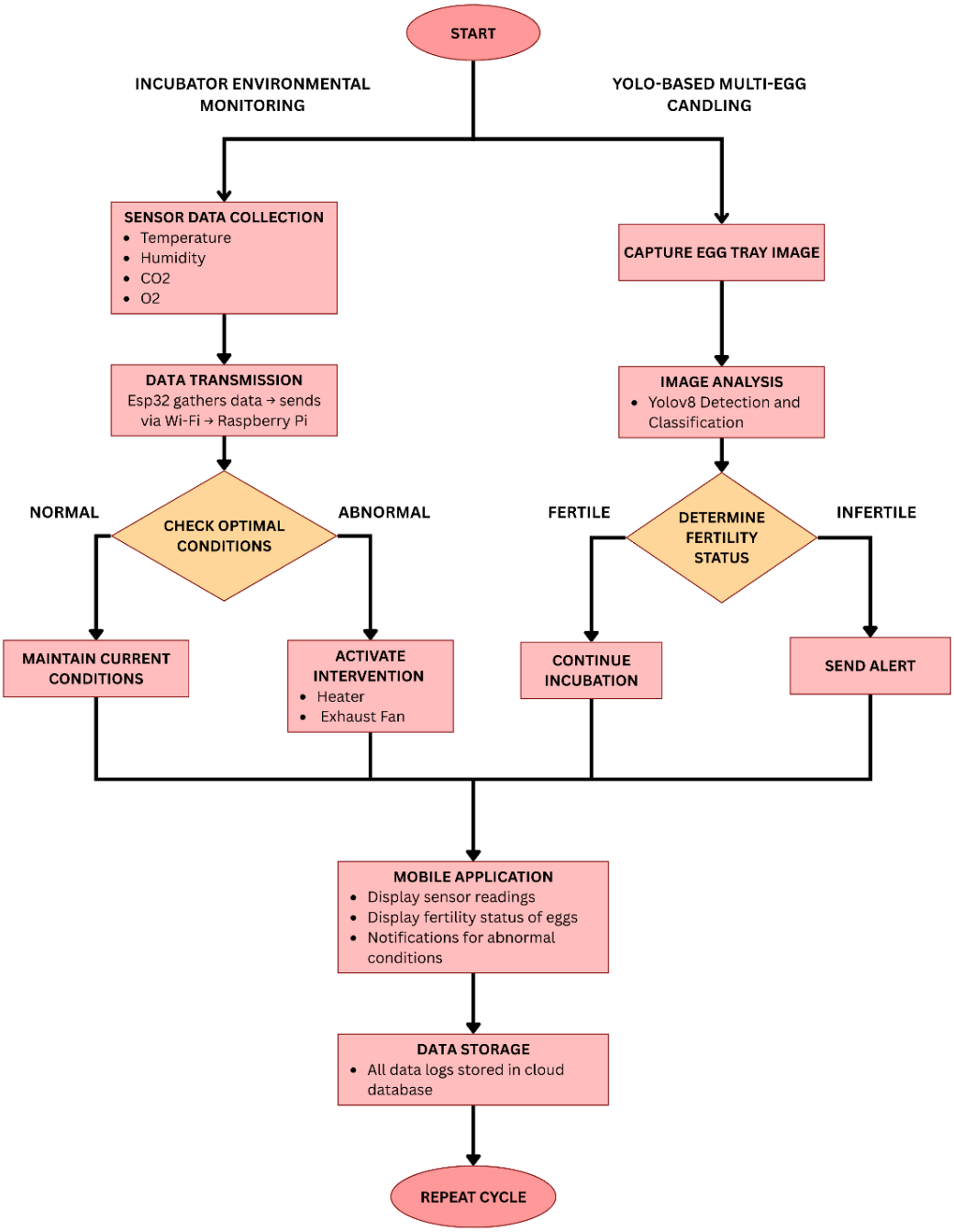


Figure 3.4. *System Process Flowchart*

Figure 3.4 illustrates the overall system process flowchart. The system begins with the initialization stage, where the incubator is powered on and all components are prepared for operation. Once active, the system simultaneously performs two primary processes: environmental monitoring and egg fertility assessment.

In the environmental monitoring process, sensor data including temperature, humidity, carbon dioxide (CO_2), and oxygen (O_2) levels are collected continuously. These data are transmitted by the ESP32 to the Raspberry Pi via Wi-Fi for processing. The system then evaluates whether the current conditions fall within the optimal incubation range. If the conditions are within the acceptable limits, the system maintains the current state. Otherwise, corrective actions are triggered, such as activating the heater or the exhaust fan to restore proper environmental conditions.

At the same time, the egg monitoring subsystem captures images of the egg tray using a camera module. These images are processed through a YOLOv8-based image analysis model to detect and classify egg fertility status. Based on the analysis, the system determines whether the eggs are developing normally. If the eggs are healthy, the incubation process continues; otherwise, the system sends alerts to notify the user of any abnormalities.

All collected data, including sensor readings and egg fertility results, are displayed through a mobile application, allowing real-time monitoring. Additionally, the data are stored in a cloud database for logging and further analysis. Finally, the entire process operates in a continuous loop, ensuring constant monitoring, control, and decision-making throughout the incubation period.

3.5.2. Phase Transition Logic

While the system process flowchart outlines the overall loop, the system also applies phase-specific logic aligned with the 21-day incubation timeline. During the setting phase (Days 1–18), the environmental monitoring branch follows thresholds optimized for embryo growth, and the mechanical turning process which is integrated into the maintain current conditions step is active.

When the system transitions to the hatching phase (Days 19–21), its control logic automatically adjusts the operational parameters suitable for the hatching phase. This adaptive logic ensures that the intervention and monitoring steps shown in Figure 3.4 remain biologically appropriate for every stage of the incubation cycle. The specific numerical threshold values and decision-making setpoints used to trigger these automated interventions are detailed further in Section 3.8 of this chapter.

3.6 Hardware Development

3.6.1 Environmental Design Parameters and Thresholds

Phase	Temperature (°C)	Humidity(%)	Egg Turning
Setting (Days 1-18)	37.5	58	Rotates every hour
Hatching (Day 19-21)	36.9	66-75	Stops on day 19

Table 3.1. *Egg Incubator Standards based on the Philippine National Standards 2023*

Table 3.1 shows the temperature and humidity standards used in managing the critical 21-day incubation cycle between two (2) distinct phases. During the setting phase (days 1-18), the environment will be maintained by the incubator at 37.5°C with 58% humidity, while an automated mechanism rotates the eggs every hour to ensure

healthy embryo development. Upon reaching the hatching phase(days 19-21), the system automatically adjusts to assist the final breakout; the temperature is slightly reduced to 36.9°C, and humidity is increased to a range of 66%–75% to soften the shells. Simultaneously, all egg-turning ceases on Day 19, allowing the chicks to orient themselves correctly for pipping. This precise, data-driven transition ensures that the environmental conditions remain optimal for every stage of the life cycle.

Table 3.2. *Optimal Gas Conditions for Egg Incubation*

Phase	**Carbon Dioxide (CO_2)**	**Oxygen (O_2)**
Setting (Days 1-18)	0.3 - 0.5%	21%
Hatching (Day 19-21)	0.75%	21%

Table 3.2 outlines the optimal atmospheric conditions required for successful egg incubation according to Mauldin (2008), emphasizing the balance between carbon dioxide (CO_2) and oxygen (O_2) levels. During the setting phase (Days 1–18), CO_2 levels are maintained between 0.3% and 0.5% to avoid developmental issues, while O_2 is kept at a constant 21%. As the embryos transition to the hatching phase (Days 19–21), the CO_2 threshold increases to 0.75% to accommodate the higher metabolic output of newly hatched chicks, while the O_2 requirement remains at 21% to prevent significant drops in hatchability.

3.6.2 Incubator Structural and Casing Design

The physical architecture of the incubator is engineered to execute the four operational pillars of the Project Design Theory, environmental control, mechanical turning, vision-based detection, and connectivity in a single, automated environment.

As shown in Figure 3.5, the enclosure is divided into a specialized upper housing and a visible incubation chamber to facilitate these processes.



Figure 3.5. Overall Incubator Design

The upper housing, located directly above the viewing window, acts as the primary innovation hub for the vision-based fertility detection process. This area provides the necessary clearance to house an overhead camera system positioned directly above the egg tray. This arrangement allows the camera to perform automated candling by capturing high-resolution images of the eggs from a top-down perspective, which are then analyzed by the YOLOv8 model without the need for manual handling. This section also serves as the protected enclosure for the system's microcontroller and wireless modules, which manage the connectivity process by transmitting real-time data to the mobile interface.

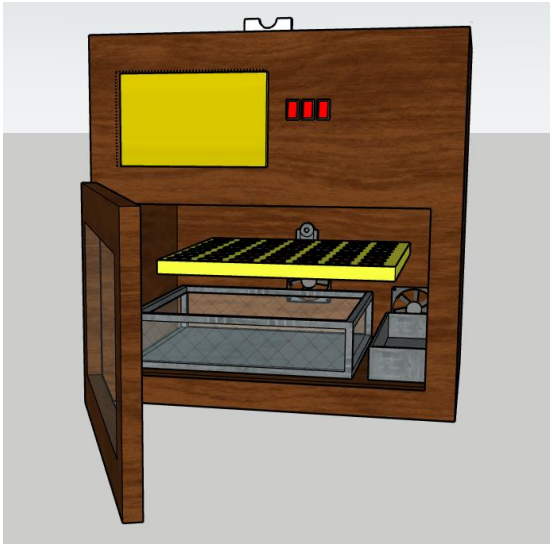


Figure 3.6. *Incubator Interior Design*

The visible incubation chamber serves as the operational site for the environmental control and mechanical processes. This chamber contains the motorized egg tray assembly, which fulfills the mechanical turning requirements. To ensure proper climate regulation, the ventilation system, including the intake vents and an integrated exhaust fan, is situated within this visible area. This allows for the precise management of air exchange and gas concentration levels as monitored by the sensor array. By placing the heating elements, hydration tray, and sensors in the lower chamber, the hardware maintains a closed-loop environment where heat and humidity are regulated in direct response to sensor feedback, ensuring the stability required for successful embryonic development.



Figure 3.7. *Top Panel*

To facilitate long-term reliability, the top of the enclosure features an integrated handle, providing manual access to the upper chamber for device maintenance and camera calibration.

3.6.3 Mechanical Turning and Vision Mounting Systems



Figure 3.8. *Overhead Camera Field of View for Automated Candling*

Figure 3.8 provides a top-down perspective, illustrating the precise field of view captured by the high-resolution camera system mounted within the incubator's upper chamber. This angle establishes the spatial relationship necessary to execute the automated vision-based fertility detection process.

From this position, the camera maintains a clear, unobstructed overlook of the main egg tray and specimen array. The yellow tray, featuring its 7x8 matrix of 56 individual egg compartments, is centered within the frame. This comprehensive coverage ensures that periodic image capture can occur simultaneously for the entire batch. The uniform illumination and fixed distance between the lens and the tray allow the specialized YOLOv8 computer vision model to analyze individual eggs during candling without requiring physical handling or manual positioning. The clear definition of each compartment visible in this view is critical for the model to register, segment, and classify the fertility status of every egg in the system efficiently.

3.6.4. System Block Diagram

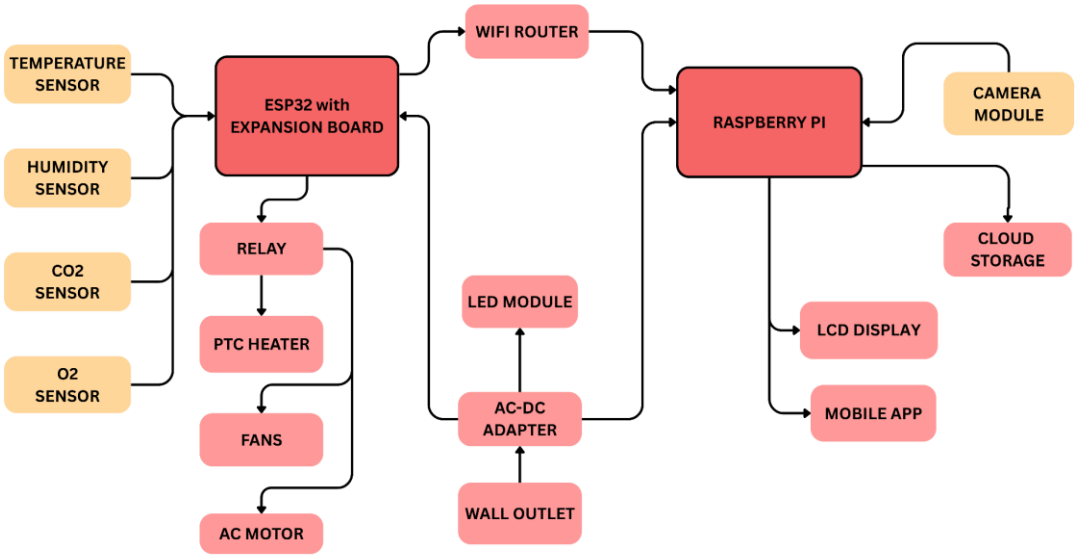


Figure 3.9 *System Block Diagram*

Figure 3.9 shows the system block diagram of the proposed Egg incubator. The system consists of an integrated hardware and software architecture designed to automate and monitor the egg incubation process. Environmental parameters, including temperature, humidity, carbon dioxide (CO_2), and oxygen (O_2) levels, are continuously measured using corresponding sensors connected to the ESP32 microcontroller with an expansion board. The ESP32 serves as the primary control unit, processing sensor data and actuating components such as the PTC heater, circulation fans, and AC motor through a relay module to maintain optimal incubation conditions. The ESP32 communicates wirelessly via a WiFi router with the Raspberry Pi, which functions as the secondary processing unit responsible for image acquisition and analysis using a connected camera module for automated egg candling. The Raspberry Pi also handles data visualization through an LCD display and transmits processed data to cloud storage, enabling remote monitoring via a mobile application. The entire system is powered by an AC wall outlet connected to an AC-DC power supply, which distributes regulated voltage to low-power components such as the ESP32 and Raspberry Pi, while high-power devices like the heater and motor are supplied separately and controlled via relays. This architecture ensures efficient system operation, stable power management, and real-time monitoring and control of incubation conditions.

3.6.5 System Component Specifications

The hardware selection for IncuVision was based on the specific needs of a chicken egg incubator and the computational requirements of real-time object detection for chicken egg fertility. The system is divided into three primary hardware groups: the sensing module, the actuator and control interface module, and the automated candling subsystem.

3.6.5.1. Component Specifications of the Sensing Module

The sensing module is designed to provide high-precision feedback on the environmental factors of the incubator, specifically, temperature, humidity, carbon dioxide, and oxygen levels.

Table 3.3. *Component Specifications of the Sensing Module*

Component	Function	Features
SCD30 Sensor	Measures CO2 concentration	NDIR Technology; I2C Interface Accuracy: ±30ppm
DHT22 Sensor	Measures temperature and relative humidity	Range: 0-100% RH Accuracy: ±2% RH
DF Robot O2 Sensor	Measures oxygen concentration	Volume: 0-25% Accuracy: ±0.15%

Table 3.3 presents the technical specifications of the sensors to be used to monitor the internal atmosphere of the incubator. These sensors were chosen for their features to ensure that the system can detect and respond to minor atmospheric fluctuations that might affect embryonic development.

3.6.5.2. Component Specifications of the Actuator and Control Module

The actuators and the control interface module allow the system to physically and automatically adjust internal incubator conditions based on the readings from the sensing module.

Table 3.4. *Component Specifications of the Actuator and Control Module*

Component	Function	Features
ESP32	The microcontroller used to run the incubation software and manage sensors and	Dual-core processor, integrated Wi-Fi and Bluetooth, low power

	controls.	consumption.
PTC Heater	A heating element to maintain optimal temperature for egg incubation.	AC; Self-regulating; high heat transfer efficiency
Exhaust Fan	For circulating air within the incubator and expelling CO2.	12V DC; 4000RPM; automated CO2 expulsion
Circulation Fan	Ensures uniform temperature and moisture distribution	AC, 3000RPM
AC Motor	Used for turning the eggs periodically for even incubation.	220V asynchronous; 2.5/3 RPM with high torque
4 Relay Module	Used for controlling the heater, led modules, and fans through the ESP32.	Signal switching, 5V Opto-isolated
20x4 LCD screen	Used for real-time farm-site monitoring	I2C Communication
12V 10A Adapter	Converts AC to DC voltage to power components.	Switching power

Table 3.4 describes the ESP32 microcontroller and the various devices it triggers, such as the heating element, ventilation fans, and the kinematic turning motor, while also specifying the relay module used to interface low-power signals with high-voltage loads.

3.6.5.3. Component Specifications for Automated Multi-Egg Candling

The automated multi-egg candling feature integrated within the incubator allows for high-precision, remote egg fertility diagnostics using computer vision. By utilizing the YOLOv8 model for real-time identification, the system aims to improve assessment accuracy compared to subjective manual methods.

Table 3.5. *Component Specifications of the Automated Multi-Egg Candling*

Component	Function	Features
Raspberry Pi 5	Serves as the main processing unit for the multi-egg candling and executes the YOLOv8 fertility detection algorithm	Quad-core, 8GB RAM, integrated GPU, and wireless connectivity.
RPI Camera Module 3	Captures images for candling and fertility detection.	12MP Resolution
LED Modules	Used for illuminating the eggs during the candling process.	High-intensity white light
12-5V Converter	Supplies stable 5V power for RPi 5	DC-DC Buck, voltage regulator

Table 3.5 outlines the hardware dedicated to the automated computer vision and diagnostic process. This module centers on the high-compute Raspberry Pi 5 unit and the high-resolution camera, supported by an integrated LED backlighting array and specialized power regulation to ensure stable operation during real-time image acquisition and AI inference.

3.6.6. Electrical System Design and Wiring Specifications

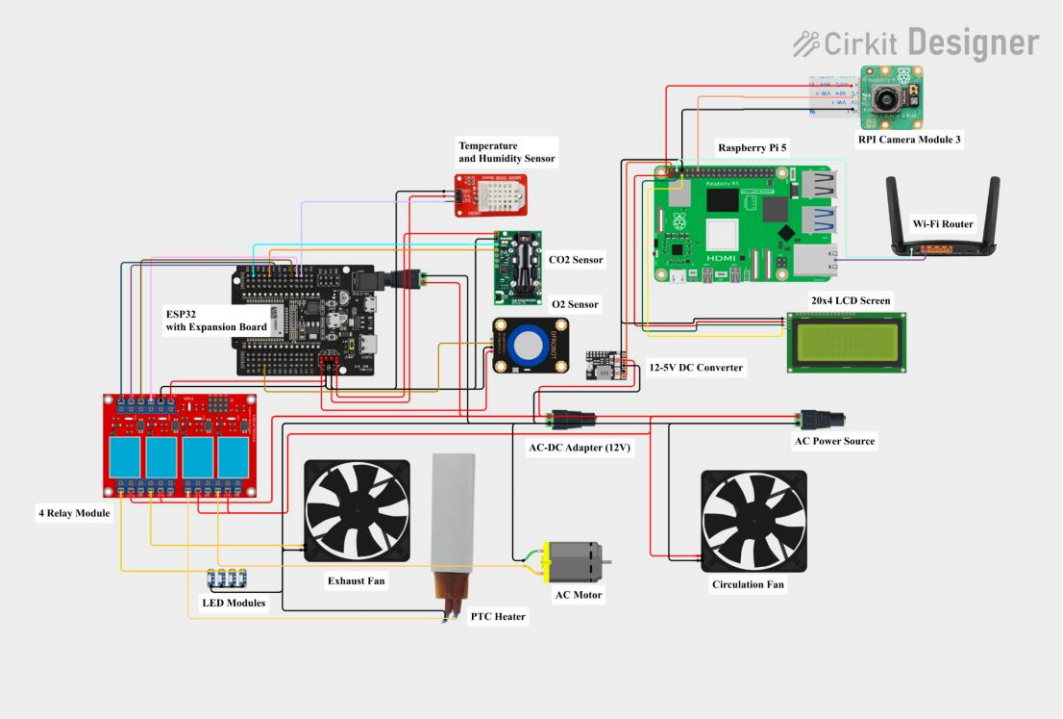


Figure 3.10. *Schematic Diagram of the IncuVision System*

Figure 3.10 depicts the overall electrical architecture, featuring the integration of a dual-controller setup with parallel operation, together with the sensing and actuation modules. The schematic details the routing of signal and power lines required to support the simultaneous operation of real-time environmental feedback loops and high-compute vision processing.

Table 3.6. *Wiring Specifications of the IncuVision System*

Component	Pin/Port	Pin Description	Power Source	Notes
SCD30 Sensor	GPIO	*I² C* Serial Data/Clock	3.3V (ESP32)	Sensor for CO2 concentration
DHT22 Sensor	GPIO	Digital Input	3.3V (ESP32)	Sensor for temperature and relative humidity
02 Sensor	GPIO	Analog Input (ADC)	5V (ESP32)	Measures oxygen concentration

Relay Channel 1	GPIO	Digital Output	5V (Rail)	Triggers exhaust fan for gas exchange.
Relay Channel 2	GPIO	Digital Output	5V (Rail)	Triggers LED Modules for candling.
Relay Channel 3	GPIO	Digital Output	5V (Rail)	Triggers PTC Heater for thermal control.
Relay Channel 4	GPIO	Digital Output	5V (Rail)	Triggers AC Motor for egg turning.
20x4 LCD Screen	GPIO (RPi)	*I² C* Serial Data/Clock	5V (RPi)	Display local real-time system status.
Camera Module 3	MIPI CSI Port	Camera Serial Interference	5V (RPi)	High-resolution input for YOLOv8 model.
Wi-Fi Router	Ethernet Port	Network Interference	12V DC	Bridges RPi and ESP32 to the cloud.
Circulation Fan	Direct to 12V	Always-ON Rail	AC power	Constant operation for thermal uniformity.
12V-5V Converter	Power Pins	Voltage Step-Down	12V (Adapter)	Regulates 5V for MCU and RPi stability.
ESP32 Controller	VIN/GND	Power Input	5V (Converter)	Master microcontroller for sensing/relays.
Raspberry Pi 5	USB-C/GPIO	Power Input	5V (Converter)	High-compute unit for vision processing.

Table 3.6 details the pin assignment and the protocol for data communication used by each hardware device present in the IncuVision system. The table constitutes

the link between the design process and the practical implementation of hardware in terms of electrical circuits.

3.6.7 System Power Consumption Analysis

Table 3.7. *Total Daily Power Consumption*

Component	Power (W)	Voltage (V)	Current (A)	Duty Cycle (%)	Hours per Day	Energy Consumption (Wh/Day)
Raspberry Pi 5 (Vision Engine)	8.00	5	1.600	100	24	192.00
ESP32 (Main Controller)	1.10	5	0.220	100	24	26.40
Wi-Fi Router	6.00	12	0.500	100	24	144.00
AC Circulation Fan (Always on)	20.00	220	0.091	100	24	480.00
PTC heater	30.00	12	2.500	25	6.0	180.00
DC Exhaust Fan	3.50	12	0.290	10	2.4	8.40
AC Turning Motor	6.00	220	0.027	1.67	0.4	2.40
20x4 LCD Display	1.20	5	0.240	100	24	28.80
LED Module Array (56pcs)	12.00	12	1.000	0.28	0.067	0.80
Integrated Sensor Array (SCD30,02,DHT22)	0.30	3.3/5	0.090	100	24	7.20
Total Energy Consumption/ Estimated kWh per Day	-	-	-	-	-	≈1,070 Wh/day; ≈ **1.07 KWh/day**

Table 3.7. Show the power consumption behavior of all integrated elements and it is calculated on the basis of their rated power, duty cycles, and hours of operation per day. Duty cycle is 100% is given to elements that run continuously and lower duty cycles are allocated to elements that can be intermittently controlled. Energy consumption of the system is computed to be 1.07 kWh/day.

3.7 YOLOv8s Model Development

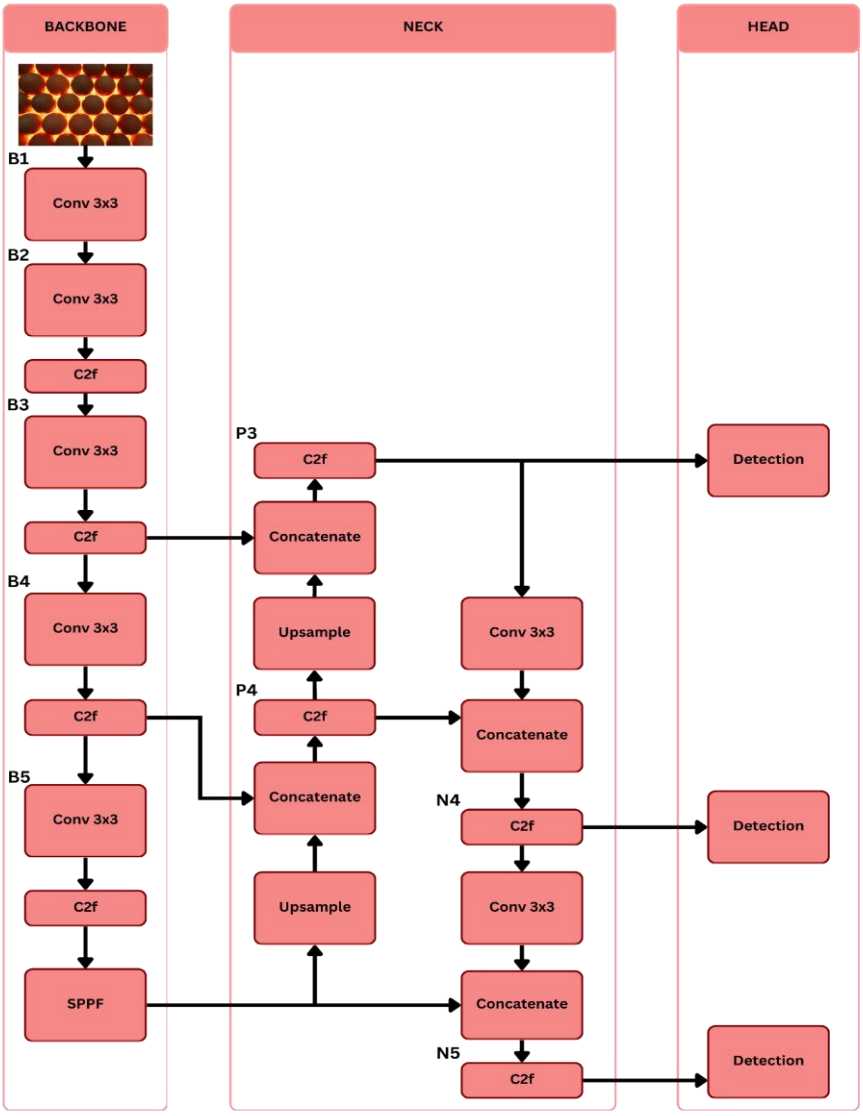


Figure 3.11 *YOLOv8s Architecture*

The YOLOv8s architecture represents an important departure from previous versions, focusing on a more streamlined and efficient approach to real-time object detection (Torres, 2024). The framework is structured into three distinct functional segments: the Backbone, the Neck, and the Head. According to documentation from YOLOv8.org, the Backbone utilizes a modified CSPDarknet53 structure, which now incorporates C2f modules in place of the older C3 blocks (Torres, 2024). This transition is critical because the C2f (Cross Stage Partial Bottleneck with 2 Convolutions) module facilitates a richer gradient flow by branching the bottleneck structures, allowing the model to maintain a lightweight profile while improving feature extraction (Torres, 2024). The backbone concludes with a Spatial Pyramid Pooling - Fast (SPPF) layer, which ensures the model can capture global context by pooling features at multiple scales.

The Neck of the architecture serves as the feature collection, utilizing a Path Aggregation Network (PANet) strategy to fuse features across different spatial resolutions. This stage is designed to merge high-level conceptual information from the deeper layers with high-resolution spatial details from the earlier stages of the backbone (Torres, 2024). By using a series of upsampling and concatenation operations, the neck ensures that the final feature maps provided to the detection heads are strong enough to identify objects of varying sizes and aspect ratios.

Finally, the Head constitutes the prediction layer of the network. A key innovation in YOLOv8 is the implementation of an anchor-free detection mechanism, which eliminates the need for predefined anchor boxes that often require manual tuning (Torres, 2024). This design utilizes a decoupled head approach, where classification and localization tasks are processed through separate branches. This separation allows the model to optimize for bounding box

accuracy and class probability independently, leading to higher precision in complex environments (Torres, 2024).

Building upon this architectural framework, the YOLOv8s variant was selected as the optimal model for the IncuVision system because it demonstrated the best overall performance in comparative testing for egg fertility (Bucag & Landicho, 2026). It outperformed other variants by achieving a superior balance of detection accuracy and localization precision, specifically reaching a mean Average Precision (mAP) of 0.943 and a Recall of 0.99.

3.7.1 Image Acquisition and Dataset Construction

The development of a robust classification model begins with the construction of a specialized dataset consisting of high-resolution egg candling images. To capture the precise biological indicators of life, data acquisition is performed across a multi-day timeline, specifically targeting the window between day 3 and day 18 of incubation where embryonic features and vascular networks become distinct. These images are captured using a localized candling setup integrated directly within the incubator, ensuring consistent lighting and focal points. To prevent model overfitting and ensure the system does not simply memorize tray positions, the eggs are shuffled periodically throughout the collection process to create a diverse range of spatial samples. This approach ensures that the AI Brain learns to identify fertility based on internal biological markers rather than external environmental factors, maintaining perfect alignment with the real-world optical conditions of the final implementation.

Table 3.8. *YOLOv8s Dataset Distribution*

Dataset Distribution for YOLOv8s	Image Quantity	Percentage(%)
1. Training Images		70%

2. Validation Images		20%
3. Testing Images		10%
Total		**100%**

As illustrated in Table 3.8., the dataset for the IncuVision fertility detection model is partitioned into three functional subsets to ensure robust performance and accurate generalization. The Training set, comprising 70% of the total image quantity, serves as the primary data source for the YOLOv8s architecture to learn the distinguishing visual features of fertile and infertile chicken embryos. A 20% Validation set is utilized during the training process to monitor for overfitting and to tune the model's hyperparameters, ensuring it can handle variations in lighting or egg orientation. Finally, the remaining 10% is reserved as a Testing set, which provides an unbiased final evaluation of the system's classification accuracy using images the model has never encountered. This standard 70-20-10 distribution ensures that the automated candling system remains both precise and reliable for real-time poultry farming applications.

3.7.2 Image Annotation and Pre-processing

Preparing the dataset for model development involves a rigorous standardization and labeling process to ensure the YOLOv8s architecture receives high-quality input data. By utilizing specialized annotation tools, bounding boxes are meticulously drawn around each egg to establish the ground truth for both fertile and infertile classifications. These raw images are then normalized to a consistent 640x640 pixel resolution to maintain compatibility with the model's input layers. To ensure the detection system can generalize across different shell textures and lighting conditions,

data augmentation techniques, such as brightness adjustments and rotations, are applied to broaden the diversity of the training samples.

3.7.3 AI Training Workflow Diagram

Table 3.9. *AI Training Workflow*

Stage	Process and Activities	Tools/ Frameworks	Desired Outcomes
1. Image Acquisition	Capturing high-resolution images of the 56-egg tray during the pilot run.	RPi Camera 3, Python	Custom dataset matching real incubator lighting.
2. Pre-processing	Resizing images to 640 × 640 and normalizing	OpenCV, Roboflow pixel values	Standardized input format for model training.
3. Data Annotation	Manual bounding box labeling of "Fertile" and "Infertile" classes.	Roboflow	Establishment of ground truth for supervised learning.
4. Data Augmentation	Applying geometric transforms (rotation, flip) and brightness variations.	Roboflow / Albumentations	Increased model robustness and dataset diversity.
5. Model Training	Iterative training of the YOLOV8s architecture for 100 times.	Google Colab, NVIDIA, GPU, PyTorch	Optimized model weights with minimized loss functions.
6. Evaluation	Performance testing using mAP, Precision, Recall, and F1-score.	YOLOv8 Val Mode, Matplotlib	Validation of classification and localization accuracy.
7. Optimization	Converting weights to an optimized format	TensorRT / Export Logic	Low-latency inference on the Raspberry Pi 5 hardware. (e.g., ONNX or FLite).

The AI training workflow for IncuVision, as detailed in Table 3.9, follows a systematic seven-stage process designed to develop a high-performance fertility detection model. The cycle begins with Image Acquisition, where high-resolution images are captured from the 56-egg tray using a Raspberry Pi Camera 3 to ensure the dataset matches real-world incubator lighting. These images then undergo pre-processing using OpenCV and Roboflow to resize them to a standardized 640x640 format and normalize pixel values. During the Data Annotation and Data Augmentation stages, manual bounding boxes are applied in Roboflow to establish the ground truth for "Fertile" and "Infertile" classes, while geometric transforms and brightness variations are introduced to increase the model's robustness and dataset diversity.

The core of the development occurs during Model Training, where the YOLOv8s architecture is iteratively trained for 100 epochs using Google Colab's NVIDIA GPU resources to minimize loss functions and optimize model weights. Following training, the system undergoes a rigorous Evaluation phase utilizing metrics such as mAP, Precision, Recall, and F1-score to validate classification and localization accuracy. Finally, the process concludes with Optimization, where the trained weights are converted into an optimized format like TensorRT, ONNX, or TFLite. This final step is crucial for achieving low-latency inference on the Raspberry Pi 5 hardware, ensuring the automated candling system operates efficiently in real-time.

3.7.4 Computer Vision Evaluation Metrics

To evaluate the developed YOLOv8 model for Egg fertility classification, the following metrics will be calculated:

a. Intersection over Union (IoU)

IoU will measure the overlap between the predicted bounding box and the ground truth bounding box, indicating localization accuracy. A higher IoU value will indicate more accurate object localization.

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union}$$

Equation 3.1 *Intersection over Union*

b. Precision

Precision will measure the proportion of correctly identified positive detections among all predicted positives. High precision indicates fewer false detections.

$$Precision = \frac{TP}{TP + FP}$$

Equation 3.2 *Precision*

Where:

TP = True Positives

FP = False Positives

c. Recall

Recall will measure the proportion of correctly identified positive cases among all actual positives. High recall indicates fewer missed detections.

$$Recall = \frac{TP}{TP + FN}$$

Equation 3.3 *Recall*

Where:

FN = False Negatives

d. F1-Score

The F1-score will be used to evaluate the balance between precision and recall. It is particularly useful in assessing the overall reliability of the model. A high F1-score indicates that the model performs well in both detecting fertility correctly and minimizing misclassifications.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Equation 3.4 *F1-Score*

e. Mean Average Precision (mAP)

Mean Average Precision (mAP) will serve as the primary evaluation metric for YOLOv8, measuring detection accuracy across all classes and confidence thresholds.

$$mAP = \frac{1}{N}\sum_{i=1}^{N} APi$$

Equation 3.5 *Mean Average Precision*

3.8 Software Development

The software architecture of IncuVision is designed to facilitate autonomous decision-making and remote data accessibility. It involves the integration of embedded C++ firmware for the ESP32, Python scripts for the vision engine on the Raspberry Pi 5, and a mobile interface for user interaction.

3.8.1 Control Logic and Embedded Firmware

The ESP32 firmware manages the real-time environmental regulation through two distinct biological phases. During the Setting Stage (Days 1–18), as shown in Figure 3.9, the logic prioritizes thermal stability at 37.5°C and hourly kinematic turning. Upon transition to the Hatching Stage (Days 19–21), illustrated in Figure 3.10, the firmware automatically deactivates the turning motor and shifts the humidity and gas exchange thresholds to facilitate shell softening and respiratory support. Both stages utilize a closed-loop feedback mechanism, continuously reading the sensor array and triggering relays to maintain homeostasis.

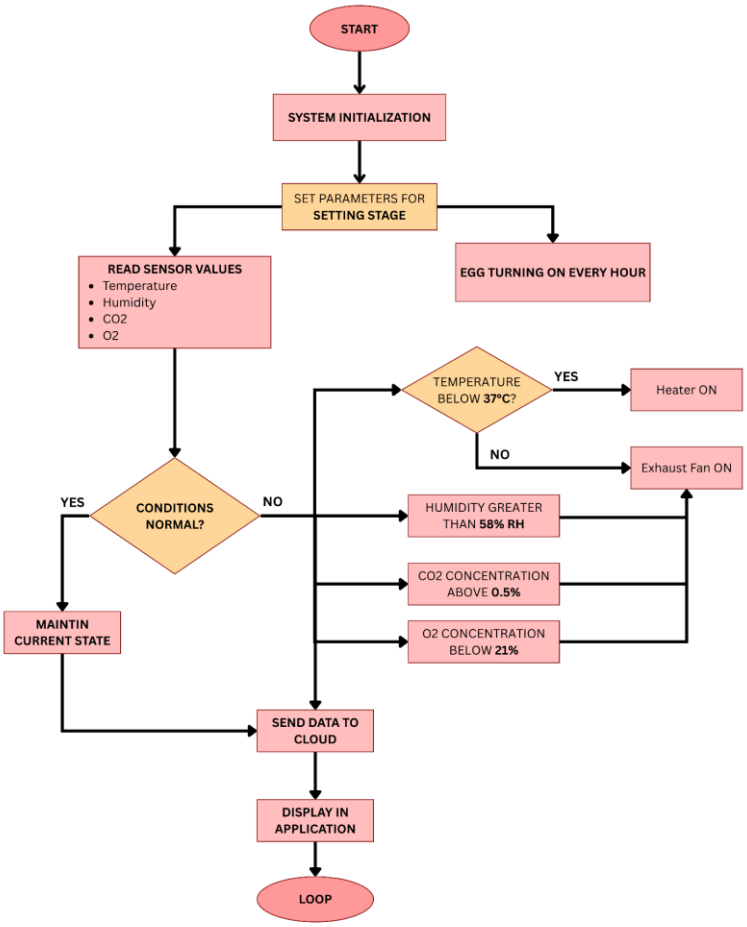


Figure 3.12 *Setting Stage Flowchart*

Figure 3.9 illustrates the control loop for the initial 18 days of incubation. The logic is centered on maintaining a high-precision thermal and gas baseline while managing the hourly kinematic modulation of the eggs to ensure steady embryonic development.

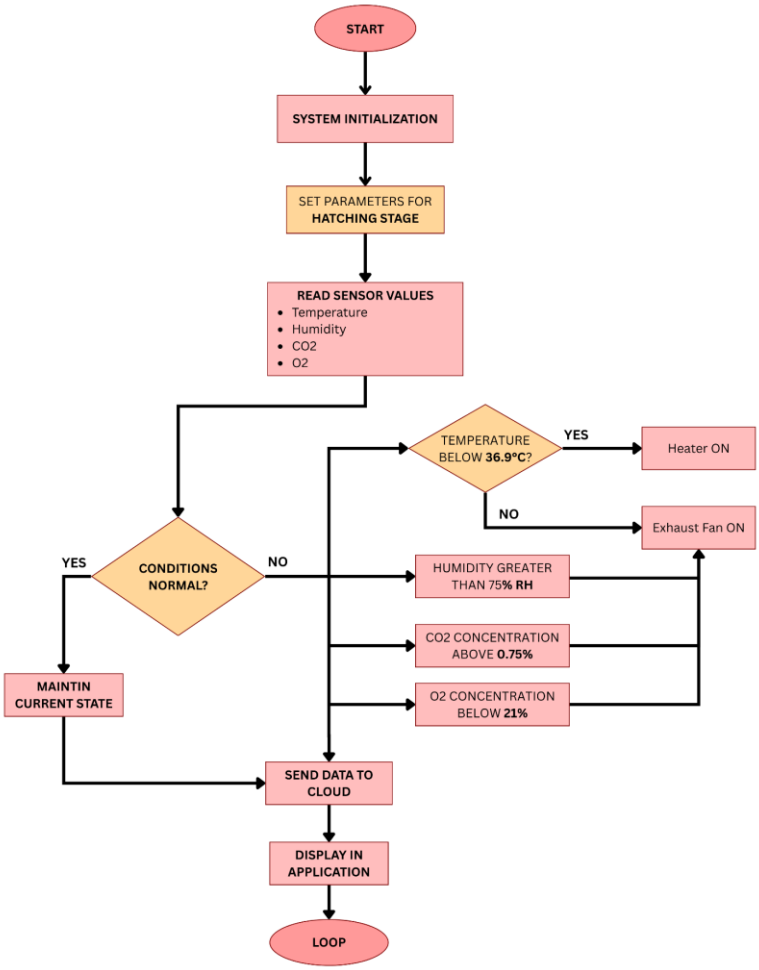


Figure 3.13 *Hatching Stage Flowchart*

Figure 3.10 details the automated logic shift for the final stage of incubation. In this phase, the system prioritizes increased moisture levels and enhanced gas exchange to support the respiratory needs of the emergent chicks while deactivating all mechanical movement to prevent physical injury during pipping.

Furthermore, the control logic includes a time-out safety mechanism to handle persistent environmental anomalies. In the event that the automated interventions, such as the activation of the heater or exhaust fan fail to restore the internal parameters to their required thresholds within a pre-defined duration, the system prioritizes biological safety by triggering a high-priority alert. This notification is instantly dispatched to the

farmer's mobile application, signaling the need for manual intervention to investigate potential hardware failures or external factors, thereby ensuring the embryos are not exposed to unstable conditions for an extended period.

3.8.2 Automated Candling and Detection Sequence

The diagnostic intelligence is executed by the Raspberry Pi 5. As detailed in the operational sequence in Figure 3.11, the system initiates a candling event based on the user-defined schedule. The software triggers the high-intensity LED array, captures a high-resolution image via the Camera Module 3, and performs pre-processing to enhance embryonic features. The YOLOv8 model then performs simultaneous multi-target detection to classify each egg as fertile or infertile. This logic ensures that non-viable eggs are identified early, allowing for timely removal to prevent internal decomposition gases from affecting the healthy batch.

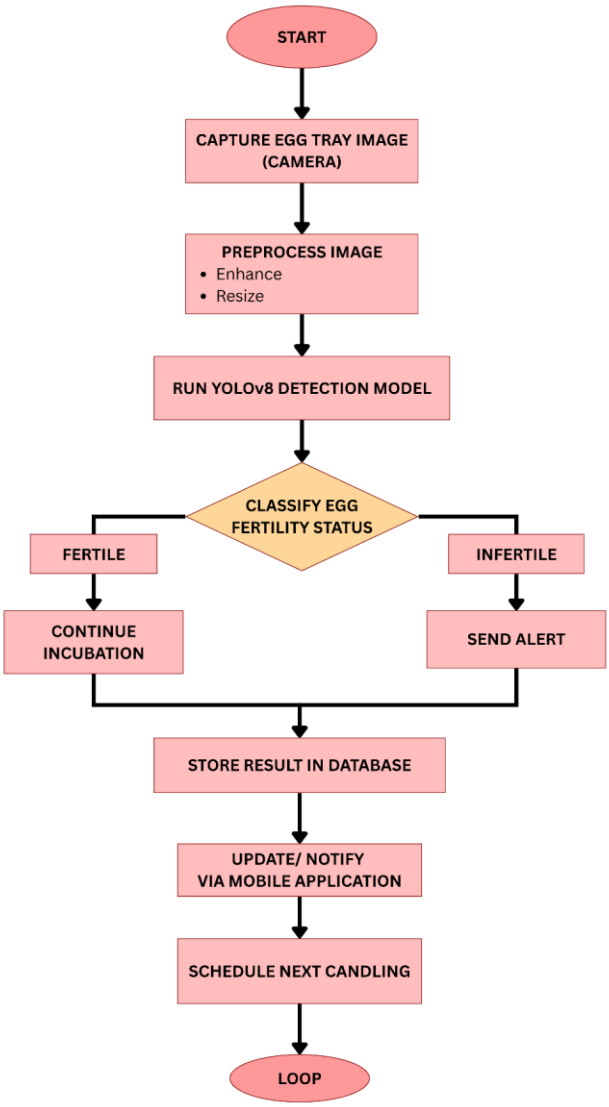


Figure 3.14 *Automated Multi-Egg Candling Flowchart*

This sequence outlines the coordination between hardware triggers and deep learning inference. It illustrates how the Raspberry Pi 5 synchronizes the illumination and acquisition hardware with the YOLOv8 model to perform instantaneous multi-target classification of the entire egg tray. The process ensures that diagnostic results are autonomously pushed to the cloud database, enabling real-time status updates and alerts on the user interface without physical interruption of the environment.

Table 3.10. *Recommended Schedule for candling*

Day	Purpose of Candling	Expected Observation
Day 7	Initial fertility check	Blood vessels or embryo shadow (fertile eggs).
Day 10	Monitoring development progress	More pronounced embryo and growth patterns.
Day 18	Final check before transitioning to hatching	Fully developed embryo or stagnant growth.

Table 3.10 summarizes the recommended schedule for egg candling. The days specified are when the eggs show distinct embryonic development during incubation.

When the system is not in the process of candling eggs, the Raspberry Pi Camera Module 3 and the high-intensity LED light source will be in a de-energized standby mode. This strategy will be adopted to limit the amount of heat produced inside the top chamber as well as any electrical interference that might occur for the atmospheric sensors. Moreover, keeping the vision system in idle mode during periods when there is no candling activity will save on the energy used by the Raspberry Pi 5.

3.8.3 Mobile Application Interface Design

To facilitate remote monitoring and enhance the system's accessibility, a dedicated mobile application will be developed using MIT App Inventor, a blocks-based visual programming environment. The application serves as the primary user interface (UI) for the poultry farmer, providing real-time visualization of the environmental and biological data collected by the ESP32 and Raspberry Pi 5.

The communication bridge is established through a cloud-based Real-time Database, which acts as a centralized repository for sensor telemetry. Environmental parameters, including ambient temperature, humidity, carbon dioxide (CO_2), and

oxygen (O_2) concentrations, are transmitted via the ESP32's Wi-Fi module at fixed intervals. Concurrently, the results of the YOLOv8 classification (fertile or infertile status) are pushed from the Raspberry Pi 5 to the cloud platform.



Figure 3.15 *MIT App Inventor interface used to develop the application*



Figure 3.16 *App Logo and Main Interface Design*

The IncuVision mobile application provides real-time monitoring of critical environmental parameters including temperature, humidity, CO_2, and oxygen concentration, along with interactive controls for connected devices such as exhaust

fans, circulation fans, and heating units. Additional screens display temperature trend graphs, egg type information, and egg turning schedules, while smart notifications alert users whenever measurements fall outside safe thresholds at any point during the incubation or hatching process.

Furthermore, the application incorporates a computer vision-based Egg Fertility Checker powered by the YOLOv8 algorithm, which analyzes captured egg images and provides accurate percentage readings of fertile and infertile eggs to guide users in planning their next candling session. Navigation throughout the app is made simple and accessible through a bottom menu bar offering direct access to the main menu, homepage, and user profile.



Figure 3.17 *Environmental Monitoring Interface*

The IncuVision mobile application serves as the primary user interface for monitoring and controlling the intelligent egg incubation system. The dashboard screen

displays an overview of the incubation progress through a countdown timer, dedicated monitoring tiles for four key environmental parameters such as Temperature, Humidity (%), Carbon Dioxide (CO_2), and Oxygen (O_2) , and toggle controls for connected hardware components including the Exhaust Fan, Circulation Fan, and PTC Heater. A temperature gauge provides a quick visual reference of the current thermal status inside the incubator.

The monitoring screen presents a real-time line graph visualizing temperature fluctuations over multiple days, along with the egg type and egg turning interval timer to ensure proper embryo development. A notification bell icon alerts users to any abnormal changes in incubation conditions. The application adopts a warm, pastel color scheme with a stylized chick logo, and features a bottom navigation bar for easy access to the Menu, Home, and Profile sections, ensuring usability for poultry farmers and breeders of varying technical backgrounds.

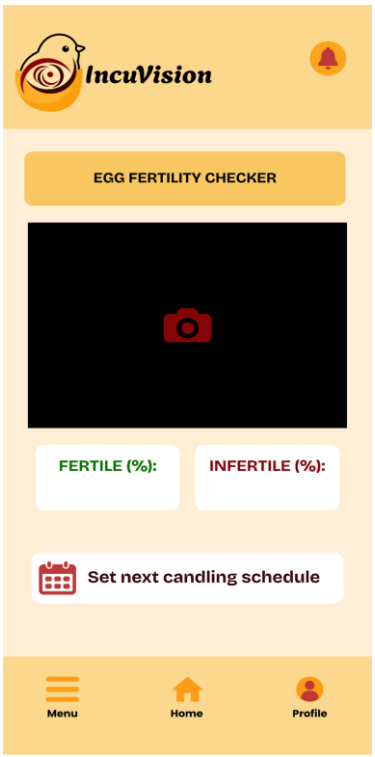


Figure 3.18 *YOLOv8 Multi-Egg Fertility Checker*

The application also includes an Egg Fertility Checker screen, which enables users to assess egg viability directly through the mobile interface. This screen features a built-in camera viewer that captures images of the egg for fertility analysis, displaying the results as percentage values for both Fertile (%) and Infertile (%) classifications. Additionally, a "Set Next Candling Schedule" button with a calendar icon allows users to plan and track upcoming candling sessions, ensuring timely monitoring of egg development throughout the incubation cycle.

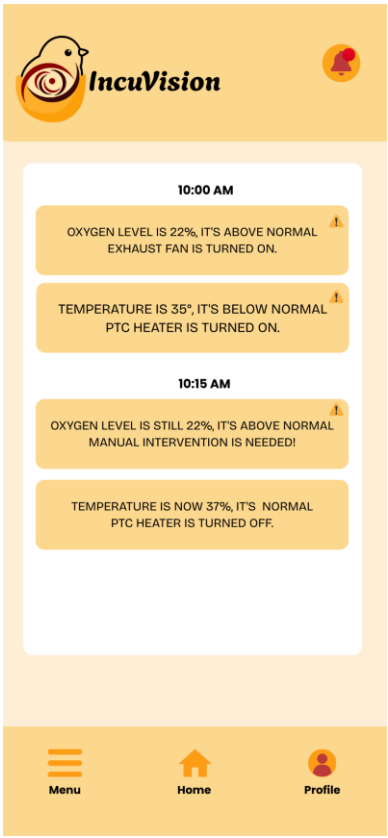


Figure 3.19 *Notification Panel*

The Notification Panel provides a timestamped log of real-time system alerts, keeping users informed of any changes or anomalies in the incubation environment. Notifications are grouped by time (e.g., 10:00 AM, 10:15 AM) and display automated status messages for critical parameters. For instance, alerts notify the user when oxygen levels exceed normal thresholds and the exhaust fan has been activated, when

temperature drops below normal and the PTC heater has been turned on, or when manual intervention is required due to persistently abnormal readings. This feature ensures that users are continuously updated on the incubator's condition, enabling prompt response to environmental irregularities without the need for constant manual monitoring.

3.9. Project Monthly Time Frame: Gantt Chart

Table. 3.11. *Project Monthly Time Frame of Major Activities/Work Plan*

Activities/Work Plan		**MAY**	**JUN**	**JUL**	**AUG**	**SEPT**	**OCT**
1.Design Phase	Finalizing system design and components						
	Developing schematic diagrams						
	Selecting materials and components						
	Incubator prototype casing						
2.Hardware and Software Development	Purchasing of devices and materials						
	Assembling and wiring the hardware						
	Programming a microcontroller (ESP32) and creating a mobile app						
	Testing hardware components						
	Building of prototype						
3.Prototype Deployment and Data Gathering	Conduction of egg incubation experiments						
	Recording temperature, humidity, and other						

	variables						
	Monitoring egg fertility (non-invasive candling)						
4. Testing and Validation	Testing automatic systems (egg turning, temperature control, etc.)						
	Validating system performance under real conditions						
	Analyzing fertility rates and hatchability						
	Final system adjustments and fine-tuning						
5. Finalization of the Research Paper							

This section outlines the projected timeline for the study, and is visually represented through the Gantt chart in Table 3.3 above. This chart serves as a strategic roadmap for the project's execution, providing a chronological breakdown of the tasks required to develop, integrate, and evaluate the IncuVision system. Through mapping these activities across the research period, the Gantt chart ensures efficient time management and confirms the feasibility of completing the multi-stage engineering and biological processes within the allotted timeframe.

3.10. Data Gathering Methods and Procedures

3.10.1. Fertility and Hatchability Assessment

The fertility and hatchability performance of eggs will be assessed to determine the effectiveness of the IncuVision system compared to a conventional incubator. Egg

batches will be subjected to similar incubation conditions using both systems for comparative analysis.

3.10.1.1 Fertility Rate Assessment

The fertility rate will be determined through a candling process, where chicken eggs will be inspected for embryonic development using the multi-egg candling system. Candling will be performed on the 7th and 18th days of incubation to assess embryo viability. The fertility rate will be computed using the formula:

$$Fertility\ Rate\ = \frac{Number\ of\ Fertile\ Eggs}{Total\ number\ of\ Eggs\ Incubated} \times 100$$

Equation 3.6. *Fertility Rate*

Table 3.12. *Fertility Rate Assessment*

Incubator Type	Total Eggs Incubated	Fertile Eggs	Fertility Rate (%)
IncuVision Prototype			
Traditional Incubator			

3.10.1.2 Hatchability Rate Assessment

Hatchability percentage will be determined at the end of the incubation period by comparing successful hatches against the total fertile eggs, calculated as follows:

$$Hatchability\ Rate\ = \frac{Number\ of\ Hatched\ Eggs}{Number\ of\ Fertile\ Eggs} \times 100$$

Equation 3.7. *Hatchability Rate*

Table 3.13. *Hatchability Rate Assessment*

Incubator Type	Fertile Eggs	Hatched Eggs	Hatchability Rate (%)
IncuVision Prototype			
Traditional Incubator			

3.10.2. Automated vs. Manual Candling Comparison

This section evaluates the performance of YOLOv8-based multi-egg candling against the traditional manual candling method in terms of accuracy and processing time. Both methods will be conducted to the same set of egg samples to ensure fair comparison.

3.10.2.1. Candling Accuracy Comparison

Accuracy of the YOLOv8-based multi-egg candling will be evaluated by comparing it to manual candling. Manual candling will serve as ground truth established through expert validation and incubation outcomes. A sample of eggs will be assessed using both methods, and accuracy rate will be computed based on the percentage of correctly identified fertile and infertile eggs.

$$Accuracy\ Rate\ = \frac{Correctly\ Identified\ Eggs}{Total\ Eggs\ Examined} \times 100$$

Equation 3.8. *Accuracy Rate*

Table 3.14. *Accuracy Comparison of Candling Methods*

Method	Total Eggs Examined	Correctly Identified Eggs	Accuracy (%)
YOLO-Based Multi-Egg			

Candling			
Manual Candling			

3.10.2.2. Time Efficiency Comparison

Time efficiency will be evaluated by recording the total time taken to candle the same sample of eggs for both methods. The average time taken per egg will also be calculated.

Table 3.15. *Time Comparison of Multi-Egg Candling and Manual Candling*

Method	Total Eggs Examined	Total Time Taken (Seconds)	Average Time per Egg (Seconds)
YOLOv8-based Multi-Egg Candling			
Manual Candling			

3.10.3. Environmental Monitoring and Stability Analysis

The environmental performance of the IncuVision system will be assessed through continuous monitoring and analysis of key incubation parameters, specifically temperature, humidity, oxygen (O2), and carbon dioxide (CO2) levels throughout the entire incubation period. The collected data will be utilized to determine the system's capability to maintain optimal environmental conditions required for proper embryo development.

3.10.3.1 Environmental Monitoring

Temperature, humidity, oxygen (O_2), and carbon dioxide (CO_2) readings will be continuously collected and recorded at predetermined intervals

throughout both the setting and hatching phases of incubation. These data will serve as the primary basis for environmental stability analysis for the proposed IncuVision system.

Table 3.16. *Environmental Monitoring Log*

Date/Time	Incubation Phase	Temperature (ºC)	Humidity (%)	Oxygen Level (%)	Carbon dioxide Level (%)

3.10.3.2 Environmental Stability Evaluation

From the environmental data recorded in Table 3.16, the mean and standard deviation of temperature, humidity, oxygen (O_2), and carbon dioxide (CO_2) levels will be calculated to evaluate the consistency of the incubator conditions throughout the incubation process. Lower standard deviation values will indicate more stable environmental regulation.

Table 3.17. *Temperature, Humidity, O_2, and CO_2 Stability During Incubation*

Environmental Parameter	Target Value	Mean Recorded Value	Standard Deviation
Temperature (°C)			
Humidity (%)			
Oxygen (O_2 %)			
Carbon Dioxide (CO_2 %)			

3.11. Data Analysis and Statistical Treatment

This study will evaluate the performance and efficiency of the system by analyzing key indicators such as fertility rate, hatchability rate, environmental stability, and overall system functionality. To process these variables, the research will employ appropriate statistical tools using the Statistical Package for Social Sciences (SPSS) version 22, which provides robust capabilities for accurate data computation and interpretation.

To accomplish this, the study will employ both descriptive and inferential statistical approaches. Descriptive statistics, specifically mean, percentage, standard deviation, and mean absolute error (MAE) will be used to organize and summarize the data gathered. For inferential analysis, various tests will be applied to determine whether observed differences between the proposed IncuVision system and the traditional incubation practices are statistically significant.

3.11.1 Fertility and Hatchability Rate Comparison

Fertility and hatchability data will be recorded according to the incubator type used for the incubation process. Descriptive statistics will be utilized to calculate the percentage of fertility rate and hatchability rate for both methods.

Furthermore, an independent samples t-test will be used to determine whether there is a significant difference in fertility and hatchability rates between eggs incubated using the IncuVision system and those incubated using traditional incubation methods. This test is appropriate because two separate groups of egg batches will be evaluated under different incubation systems. The analysis will be conducted using a significance level of $p \leq 0.05$ to confirm whether there is a significant difference between the two incubation methods.

$$t = \frac{\underline{x_1} - \underline{x_2}}{\sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}}$$

Equation 3.9. *Independent Samples T-test*

Where:

$\underline{x_1}$ = mean fertility/hatchability rate of IncuVision

$\underline{x_2}$ = mean fertility/hatchability rate of traditional incubation

s_1, s_2= variances of the two groups

n_1, n_2= sample sizes of the two groups

The following hypotheses will be tested:

Null Hypothesis (H_{01}): There is no significant difference in fertility and hatchability rates between IncuVision and traditional incubation methods.

Alternative Hypothesis (H_{11}): There is a significant difference in fertility and hatchability rates between IncuVision and traditional incubation methods.

3.11.2. Accuracy Comparison of YOLO-Based Multi-Egg Candling vs Multi-Egg Candling

McNemar's test will be utilized to determine whether there is a significant difference in classification accuracy between YOLOv8-based multi-egg candling and manual candling. This test is appropriate for paired categorical data using the same set of egg samples. The analysis will be conducted using a significance level of $p \leq 0.05$ to confirm whether there is a significant difference between the accuracy of the two candling methods.

$$x^2 = \frac{(|b - c| - 1)^2}{b + c}$$

Equation 3.10. *McNemar's Test*

Where:

b = number of samples incorrectly classified by manual candling but correctly classified by YOLOv8

c = number of samples correctly classified by manual candling but incorrectly classified by YOLOv8

The following hypotheses will be tested:

Null Hypothesis (H_{02}): There is no significant difference in classification accuracy between YOLOv8-based candling and manual candling.

Alternative Hypothesis (H_{12}): There is a significant difference in classification accuracy between YOLOv8-based candling and manual candling.

3.11.3 Time Efficiency Comparison of YOLO-Based Multi-Egg Candling vs Multi-Egg Candling

The processing time for individual eggs and entire batches will be documented for both candling techniques. Descriptive statistics will be initially used to calculate the average duration required for each method.

Subsequently, a **paired t-test** will be performed to compare the mean values and determine if there is a statistically significant difference in time efficiency between the YOLOv8-based multi-egg candling and manual candling approaches. This statistical test is appropriate as both methods will be applied to the same set of egg samples, resulting in paired observations for time comparison. A p-value of less than or equal to 0.05 ($p \leq 0.05$) will be used as the threshold to confirm the statistical significance between the two candling methods.

$$t = \frac{\underline{d}}{Sd/\sqrt{n}}$$

Equation 3.11. *Paired T-test*

Where:

$\underline{\boldsymbol{d}}$ = mean difference between paired observations

$\boldsymbol{Sd}$ = standard deviation of paired differences

$\boldsymbol{n}$ = number of paired samples

The following hypotheses will be tested:

Null Hypothesis (H_{03}): There is no significant difference in time efficiency between YOLOv8-based candling and manual candling.

Alternative Hypothesis (H_{13}): There is a significant difference in time efficiency between YOLOv8-based candling and manual candling.

3.11.4 Environmental Stability Analysis

The stability of temperature, humidity, oxygen (O_2), and carbon dioxide (CO_2) levels will be evaluated using descriptive statistics, specifically mean and standard deviation to summarize sensor data. Furthermore, **Mean Absolute Error (MAE)** will be calculated to measure the average deviation of each environmental parameter from their respective optimal incubation thresholds. Lower MAE values will indicate that the proposed incubator has sufficient environmental control and higher system stability.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|x_i - t|$$

Equation 3.12. *Mean Absolute Error*

Where:

x_i = recorded environmental value

t = target threshold value

n = total number of observations

3.12 Emergency Response Plan

Given the nature of the experimental activities conducted at the selected poultry farm in Zamboanga City, this section presents the Emergency Response Plan (ERP) for the IncuVision study. This outlines procedures for managing equipment, electrical, biological, and physical hazards that may occur during the IncuVision study at the selected poultry farm. All researchers must review the plan before fieldwork and keep a printed copy available on-site.

3.12.1 Preventive Measures

To minimize the likelihood of emergencies, the following preventive protocols shall be observed prior to and during each farm visit:

To reduce the risk of emergencies, researchers shall:

- Inspect the IncuVision system and wiring before operation.
- Wear appropriate PPE (gloves, face mask, and closed-toe shoes).
- Handle eggs properly and dispose of rotten or infertile eggs according to farm protocols.
- Monitor the IncuVision mobile application for alerts at least twice daily.
- Maintain proper ventilation around the incubator.
- Disconnect power before conducting inspections or repairs.
- Ensure the farm manager or designated personnel is informed of the day's activities.

3.12.2 Emergency Response Procedures

Table 3.18 presents the identified emergency scenarios, their associated risks, the prescribed immediate responses, and the responsible persons or follow-up actions for each scenario.

1503 **Table 3.18** *Emergency Response Procedures for the IncuVision Experimental Phase*

Emergency Scenario	Potential Risk	Immediate Response	Responsible Person / Follow-up
Power outage or electrical failure	Loss of incubation conditions; embryo death due to temperature drop	Cut power at source. Manually monitor temperature. Log the outage time and duration.	Lead researcher on duty; notify research adviser. Assess embryo viability before resuming.
PTC heater or component overheating	Fire hazard; incubation environment disrupted	Shut down the system immediately. Ventilate the area. Allow components to cool before inspection.	Lead researcher; replace or repair faulty component before restarting. Log the incident.
AC motor failure (egg-turning stops)	Embryonic mortality from static egg position	Manually turn eggs every four hours. Log each manual turning with timestamp.	Any researcher present; acquire replacement motor and resume automation once resolved.
Detection of rotten or decomposing egg (gas odor or visible leakage)	Toxic gas release; risk of egg explosion; batch contamination	Open incubator with ventilation. Remove affected egg immediately using gloves and a sealed container.	Lead researcher with PPE (gloves, N95 mask). Disinfect the tray. Re-candle adjacent eggs.
Researcher shows symptoms of illness linked to farm exposure	Zoonotic infection or allergic reaction	Remove researcher from the farm. Wash exposed skin and eyes with water. Provide first aid.	Any researcher present; seek medical attention. Report to research adviser and university health office.
Suspected avian disease outbreak on the farm premises	Zoonotic transmission risk; experimental eggs compromised	Immediately suspend all research activities. All researchers must leave the farm.	Research adviser and farm manager; report to Bureau of Animal Industry. Do not resume until farm is cleared.
Researcher sustains injury on the farm (slip, fall, or animal contact)	Minor to serious physical injury	Apply first aid. Do not move the person if injury is severe.	Nearest researcher and farm first aider; transport to nearest clinic or hospital if needed. File an incident report.
Fire or smoke detected on the farm or near the incubator	Injury and property damage	Activate alarm and evacuate all persons immediately. Do not	All researchers; call BFP emergency line (911). Account for all team

Emergency Scenario	Potential Risk	Immediate Response	Responsible Person / Follow-up
		retrieve equipment unless safe.	members at the assembly point.
Severe weather (flooding, heavy storm) preventing safe access to the farm	Researcher safety; monitoring gap in incubation data	Suspend travel to the farm. Monitor incubator remotely via the IncuVision mobile application.	Research team (off-site) and farm manager (on-site); resume only when conditions are declared safe.

3.12.3 Incident Reporting

All emergencies or near-miss incidents must be documented and reported within 24 hours. Reports should include the event details, persons involved, actions taken, and outcomes. Any injury, disease exposure, fire, or major equipment damage must be reported to the research adviser on the same day. The research team shall also determine whether the incident affects the validity or continuation of the study.

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